

YOUNG READERS

Atom Mystery



CHARLES COOMBS

YOUNG READERS

Atom Mystery



CHARLES COOMBS

The Project Gutenberg eBook of Atom Mystery [Young Atom Detective]

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Atom Mystery [Young Atom Detective]

Author: Charles Ira Coombs

Illustrator: G. Dean Lewis

Release date: October 13, 2016 [eBook #53269]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/53269

Credits: Produced by Stephen Hutcheson, MFR and the Online
Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK ATOM
MYSTERY [YOUNG ATOM DETECTIVE] ***

YOUNG READERS
Atom Mystery
[YOUNG ATOM DETECTIVE]

By CHARLES COOMBS

ILLUSTRATED BY
G. DEAN LEWIS



Publishers GROSSET & DUNLAP, INC. *New York*

©1958 BY LANTERN PRESS, INC.
Under the title: "YOUNG ATOM DETECTIVE"
BY ARRANGEMENT WITH LANTERN PRESS, INC.
MANUFACTURED IN THE UNITED STATES OF AMERICA



CONTENTS

<u>CHAPTER ONE</u>	11
<u>CHAPTER TWO</u>	27
<u>CHAPTER THREE</u>	45
<u>CHAPTER FOUR</u>	58
<u>CHAPTER FIVE</u>	72
<u>CHAPTER SIX</u>	89
<u>CHAPTER SEVEN</u>	107
<u>CHAPTER EIGHT</u>	117
<u>CHAPTER NINE</u>	135
<u>CHAPTER TEN</u>	151
<u>CHAPTER ELEVEN</u>	161
<u>CHAPTER TWELVE</u>	172



ILLUSTRATIONS

	Page
<u><i>“Hi, there,” Eddie greeted, “Any luck?”</i></u>	51
<u><i>“Come on, Eddie, let’s go back.”</i></u>	104
<u><i>The cylinder was simple to locate.</i></u>	137
<u><i>... he saw the small rubber boat moving in.</i></u>	179

YOUNG READERS **Atom Mystery**



CHAPTER ONE

It was only a dream. Eddie Taylor would like to have finished it, but the bar of morning sunlight poking in under the window shade pried his eyes open. The dream fled. Eddie kicked off the sheet, swung his feet to the floor, and groped under the bed for his tennis shoes.

He heard his father's heavy footsteps in the hallway. They stopped outside of his bedroom door.

"You awake, Eddie?"

"I'm awake, Dad," Eddie answered.

"Breakfast's ready. Get washed and dressed."

"Be right there," Eddie said. Then, remembering the dream, he added, "Oh, Dad, is it all right if I use the Geiger counter today?"

12

Mr. Taylor opened the door. He was a big man, broad-shouldered and still thin-waisted. Eddie found it easy to believe the stories he had heard about his father being an outstanding football player in his time. Even his glasses and the gray hair at his temples didn't add much age, although Eddie knew it had been eighteen years since his father had played his last game of college football.

"You may use the Geiger counter any time you want, Eddie," Mr. Taylor said, "as long as you take good care of it. You figured out where you can find some uranium ore?"

Eddie smiled sheepishly. “I—I had a dream,” he said. “Plain as day. It was out on Cedar Point. I was walking along over some rocks. Suddenly the Geiger counter began clicking like everything.”

“Cedar Point?” his father asked. “I’ve never been out there. But, from what I hear, there are plenty of rock formations. Might be worth a try, at that. You never can tell where you might strike some radioactivity.”

13

“Do you believe in dreams, Dad?”

“Well, now, that’s a tough question, son. I can’t say that I really do. Still, one clue is as good as another when it comes to hunting uranium ore, I guess. But right now we’d better get out to breakfast before your mother scalps us. Hurry it up.” His father turned and went back down the hallway toward the kitchen.

Eddie pulled on his trousers and T shirt and went into the bathroom. He washed hurriedly, knowing that even if he missed a spot or two, he was fairly safe. During the summer months his freckles got so thick and dark that it would take a magnifying glass to detect any small smudges of dirt hiding among them. He plastered some water on his dark-red hair, pushed a comb through it, and shrugged as it snapped back almost to its original position. Oh, well, he had tried.

He grinned into the mirror, reached a finger into his mouth, and unhooked the small rubber bands from his tooth braces. He dropped them into the waste basket. He’d put fresh ones in after breakfast.

14

He brushed his teeth carefully, taking particular pains around the metal braces. The tooth-straightening orthodontist had warned him about letting food gather around the metal clamps. It could start cavities.

Finished, Eddie went out to breakfast.

“Good morning, dear,” his mother greeted him, handing him a plate of eggs.

“Hi, Mom,” Eddie said. “Gotta hurry. Big day today.”

“So your father says. But I’m afraid your big day will have to start with sorting out and tying up those newspapers and magazines that have been collecting in the garage.”

“Aw, Mom—”

“Eddie, I asked you to do it three days ago. Remember? And the Goodwill truck comes around today.”

“But, Mom—”

“No arguments, son,” his father put in calmly but firmly. “School vacation doesn’t mean that your chores around here are on vacation, too. Get at it right away, and you’ll still have time to hunt your uranium.”

“Well,” Mr. Taylor added, excusing himself from the table, “I’d better be getting over to school. I’m expecting to receive shipment of a new radioisotope today.”

The very word excited Eddie. In fact, anything having to do with atomic science excited him. He knew something about isotopes—pronounced *eye-suh-tope*. You couldn’t have a father who was head of the atomic-science department at Oceanview College without picking up a little knowledge along the way. Eddie knew that a radioisotope was a material which had been “cooked” in an atomic reactor until it was “hot” with radioactivity. When carefully controlled, the radiation stored up in such isotopes was used in many beneficial ways.

“Why don’t college professors get summer vacations, too?” Eddie asked. One reason for asking that particular question was to keep from prying deeper into the subject of the radioisotope. Much of his father’s work at Oceanview College was of a secret nature. Eddie had learned not to ask questions about it. His father usually volunteered any information he wanted known, so Eddie stuck to questions which could and would be answered.

“We get vacations,” his father said. “But—well, my work is a little different, you know. At the speed atomic science is moving today, we simply can’t afford to waste time. But don’t worry. We’ll take a week or so off before school starts in the fall. Maybe head for the mountains with our tent and sleeping bags.”

“And Geiger counter?” Eddie asked eagerly.

“Wouldn’t think of leaving it home,” his father said, smiling. “By the way, I put new batteries in it the other day. Take it easy on them. Remember to switch it off when you’re not actually using it.”

“I will,” Eddie promised. He had forgotten several times before, weakening the batteries.

It took Eddie over an hour to sort out the newspapers and magazines in the garage, tie them in neat bundles, and place them out on the front curb for the Goodwill pickup. By that time the sun was high overhead. It had driven off the coolness which the ocean air had provided during the earlier hours.

“Anything else, Mom?” he asked, returning to the house and getting the Geiger counter out of the closet. He edged toward the back door before his mother had much time to think of something more for him to do.

“I guess not, dear,” Mrs. Taylor said, smiling over his hasty retreat. “What are you going to do?”

“Think I’ll do a little prospecting,” Eddie said.

“Where?”

“Probably in the hills beyond the college,” Eddie said. The more he thought about it, the more he realized it was a little late in the day to go to Cedar Point. The best way to get there was by rowboat across Moon Bay, and that was too long a row to be starting now. Besides, there were plenty of other places around the outskirts of Oceanview where likely looking rock formations invited search with a Geiger counter.

“Are you going alone?” his mother asked.

18

“Oh, guess I’ll stop by and see if Teena wants to go,” Eddie answered casually. He tried to make it sound as though he would be doing Teena Ross a big favor. After all, she was only a girl. Eddie didn’t figure a girl would make a very good uranium prospecting partner, but most of the fellows he knew were away at camp, or vacationing with their folks, or something like that.

“She’ll enjoy it, I’m sure,” his mother said.

“I’ll take Sandy, too,” Eddie said. “He needs the exercise.”

“That’s a good idea, dear. Be back in time for an early dinner.”

Eddie let Sandy off his chain. The taffy-colored cocker spaniel yipped wildly over his freedom, racing back and forth as Eddie started down the street.

Christina Ross—whom everybody called Teena—lived at the far end of the block. Eddie went around to the side door of the

19

light-green stucco house and knocked.

“Oh, hi, Eddie,” Teena greeted him, appearing at the screen door. “I was hoping you’d come over.”

“Well, I—I just happened to be going by,” Eddie said. “Thought you might want to watch me do a little prospecting with the Geiger counter. But maybe you’re too busy.”

That’s how to handle it, Eddie thought. Don’t act anxious. Let Teena be anxious. Then maybe she’ll even offer to bring along a couple of sandwiches or some fruit.

“Oh, I’d love to go,” Teena said eagerly, “but I’m just finishing the dishes. Come on in.”

“I’m in kind of a hurry.”

“I’ll only be a minute.” She pushed the screen door open for him. “I’ll make us some sandwiches.”

“Stay here, Sandy,” Eddie said. “Sit.” The dog minded, although he looked a bit rebellious.

Eddie went inside and followed Teena to the kitchen. He felt triumphant about the sandwiches.

Teena tossed him a dish towel. “You dry them,” she said.

“Who, me?”

“Why not? You’re in a hurry, aren’t you? I can make the sandwiches while you dry the silverware.” She smiled, putting tiny crinkles in her small, slightly upturned nose. She wore her hair in a pony tail. Even though her hair was blond all year long, it seemed even lighter in the summer. Eddie couldn’t tell whether the sun had faded it, or whether her deep summer tan simply made her hair look lighter by contrast. Maybe both.

“Hello, Eddie,” Mrs. Ross said, coming into the kitchen. “Looks like Teena put you to work.”

“She always does, Mrs. Ross,” Eddie said, pretending great injury. “Don’t know why I keep coming over here.”

“I know,” Teena spoke up quickly. “It’s because we’re friends, that’s why.”

Eddie knew she was right. They were friends—good friends. They had been ever since Eddie’s family had moved to Oceanview and his father had become head of the college’s atomic-science department. In fact, their parents were close friends, also. Teena’s father was chief engineer for the Acme Aviation Company, one of the coast town’s largest manufacturing concerns.

21

“Well, I’ll be glad to finish them, Eddie,” Mrs. Ross offered. “I know how boys detest doing dishes.”

“Oh, I don’t really mind, Mrs. Ross,” Eddie said. “Besides, Teena’s making sandwiches to take with us.”

“Another prospecting trip?” Teena’s mother glanced at the Geiger counter which Eddie had set carefully on the dinette table.

“I still think there must be some uranium around here,” Eddie insisted. “And we can find it if anyone can.”

“I agree,” Mrs. Ross said. “But even if you don’t find it, you both seem to enjoy your hikes.”

“Oh, yes, it’s fun, Mother,” Teena replied, wrapping wax paper around a sandwich. “Guess I’m ready. I’ve got a bone for Sandy, too.”

22

“Don’t go too far out from town,” Mrs. Ross cautioned, as Eddie picked up the Geiger counter. “And stick near the main roads. You know the rules.”

“We sure do, Mrs. Ross,” Eddie assured her. “And we’ll be back early.”

They walked past the college campus, and toward the rocky foothills beyond. At various rock mounds and outcroppings, Eddie switched on the Geiger counter. The needle of the dial on the black box wavered slightly. A slow clicking came through the earphones, but Eddie knew these indicated no more than a normal background count. There were slight traces of radioactivity in almost all earth or rocks. It was in the air itself, caused by mysterious and ever-present cosmic rays, so there was always a mild background count when the Geiger counter was turned on; but to mean anything, the needle had to jump far ahead on the gauge, and the clicking through the earphones had to speed up until it sounded almost like bacon frying in a hot skillet.

There was none of that today. After they had hiked and searched most of the forenoon, Eddie said, “We might as well call it a day, Teena. Doesn’t seem to be anything out here.”

23

“It’s all right with me,” Teena agreed, plucking foxtails from Sandy’s ears. “Pretty hot, anyway. Let’s eat our sandwiches and go back home.”

“All right,” Eddie said. “You know, one of these days I’d like to go out to Cedar Point and scout around. Maybe we’ll find something there.” Then he told Teena about his dream.

Teena smiled. “A dream sure isn’t much to go on,” she said, “but they say it’s pretty out on Cedar Point. I’ll go any time you want to, Eddie.” She handed him one of the sandwiches.

It was midafternoon by the time they arrived back at Teena's house. They worked a while on a new jigsaw puzzle Teena had received on a recent birthday. Then Eddie said good-by and went on down the street toward his own home.

After putting Sandy on his long chain and filling his water dish, Eddie went in the back door. He put the Geiger counter in the closet and went into the kitchen.

24

"What's for dinner, Mom?" he asked.

Mrs. Taylor turned from the sink. Eddie knew at once, just seeing the expression on his mother's face, that something was wrong.

"Dinner?" his mother said absently. "It's not quite four o'clock yet, Eddie. Besides, dinner may be a little late today."

"But this morning you said it would be early," Eddie reminded her, puzzled.

"This morning I didn't know what might happen."

Then Eddie heard the sound of his father's voice coming from the den. There was a strange urgent tone in it. The door to the den was open. Eddie went through the dining room and glanced into the den. His father sat stiffly behind his homemade desk, talking rapidly into the telephone. Eddie caught only the last few sketchy words. Then his father placed the telephone in its cradle, glanced up, and saw Eddie.

25

If there had been even the slightest doubt in Eddie's mind about something being wrong, it vanished now. Mr. Taylor looked years older than he had that very morning. Worry lay deep in his eyes. He fumbled thoughtfully with a pencil, turning it end over end on his desk.

“Hello, son,” he said. He didn’t even ask whether Eddie had discovered any uranium ore that day. Always before, he had shown genuine interest in Eddie’s prospecting trips.

“Dad,” Eddie said anxiously, “what—what’s the matter?”

“It shows that much, does it, son?” his father said tiredly.

“What’s wrong, Dad?” Eddie prompted. “Or can’t you tell me?”

Mr. Taylor leaned back. “Quite a bit’s wrong, Eddie,” he said, “and I guess there’s no reason why I shouldn’t tell you. It’ll be in the evening papers, anyway.”

“Evening papers?”

26

“Eddie, you remember me mentioning this morning about that radioisotope shipment I was expecting today?”

“I remember,” Eddie said. “Did it come?”

“It did—and it didn’t,” his father said.

“What does that mean, Dad?” Eddie asked, puzzled.

“The delivery truck arrived at the school with it,” his father explained, “but while the driver was inquiring where to put it, the container disappeared.”

“Disappeared?”

“The radioisotope was stolen, Eddie,” his father said slowly. “Stolen right out from under our noses!”

27



CHAPTER TWO

At the moment, Eddie didn't pry for further information on the theft of the valuable radioactive isotope. His father had plenty on his mind, as it was. The main information was in the evening *Globe*, which Eddie rushed out to get as soon as he heard it plop onto the front porch.

He took the newspaper to his father to read first. After having finished, Mr. Taylor handed the paper to Eddie and leaned back thoughtfully in his chair.

"They've got it pretty straight, at that," Mr. Taylor said, "but I'm afraid this is going to stir up quite a bit of trouble."

28

"It wasn't your fault, was it, Dad?" Eddie defended.

"It was as much mine as anybody's, son," his father said. "Probably more so. After all, I am head of the department. I knew about the shipment. That should make it my responsibility to see that it was properly received and placed in our atomic-materials storage vault. But there is little point in trying to place the blame on anyone. I'm willing to accept that part of it. The important thing is that we recover that radioisotope. Not only is it of a secret nature, but it is also dangerously radioactive if improperly handled."

"But—but wasn't it in a safe container?" Eddie asked.

"Of course," his father said. "There were only two ounces of it in a fifty-pound lead capsule. As long as it remains in that capsule it's safe. As you know, the lead prevents any radiation

29

from escaping. Out of that capsule, however, those two ounces of radioisotope can be very dangerous.”

“Fifty pounds,” Eddie said thoughtfully. “That’s a pretty big thing to steal, isn’t it?”

“Not when it’s lead, son,” his father replied. “Not much bigger than a two-quart milk bottle, in fact.”

“Even at that, no kid could have taken it,” Eddie said.

“Kid?” His father smiled thinly. “We don’t think it was any kid, Eddie. Not by a long shot. The whole thing was carefully planned and carefully carried out. It was not the work of amateurs.”

Eddie read the newspaper account. The small truck from Drake Ridge, where one of the country’s newest atomic reactors was located, had arrived earlier than expected at Oceanview College. It had backed up to the receiving dock where all of the college supplies were delivered. Since deliveries during vacation months were few, there was no one on the dock when the truck arrived. A half hour later, when the delivery was expected, there would have been. The truck’s early arrival had caught them unprepared.

The driver had left the truck and had gone around the building to the front office. It had taken him less than five minutes to locate the receiving-dock foreman. Together, they had returned through the small warehouse and opened the rear door onto the dock.

During that short time someone had pried open the heavy padlock on the delivery truck’s rear door and had stolen the fifty-pound lead capsule containing the radioisotope.

Dusty footprints on the pavement around the rear of the truck indicated that two men had carried out the theft. A heavy iron

pry bar had been dropped at the rear of the truck after the lock was sprung. It was a common type used by carpenters. There were no fingerprints or other identifying marks on it. The footprints were barely visible and of no help other than to indicate that two men were involved in the crime.

“Dad,” Eddie asked, looking up from the paper, “how could anyone carry away something weighing fifty pounds without being noticed?”

31

“Chances are they had their car parked nearby,” his father said. “As you know, there are no fences or gates around Oceanview College. People come and go as they please. As a matter of fact, there are always quite a few automobiles parked around the shipping and receiving building, and parking space is scarce even during summer sessions. Anyone could park and wait there unnoticed. Or they could walk around without attracting any undue attention.”

“But, Dad,” Eddie continued, “how would the men know that the delivery truck would arrive a half hour early?”

“They wouldn’t,” his father said. “They may have had another plan. The way things worked out, they didn’t need to use it. The early delivery and the business of leaving the truck unguarded for a few minutes probably gave them a better opportunity than they had expected. At least, they took quick advantage of it.”

“I don’t see what anyone would want with a radioisotope,” Eddie said. “Maybe they figured there was something else inside of that lead capsule.”

32

“That’s unlikely, son,” Mr. Taylor said. “Believe me, it was no common theft. Nor were the thieves ordinary thieves. That isotope was a new one. A very secret one. Our job at the college was to conduct various tests with it in order to find out

exactly how it could best be put to use as a cure for disease, or for sterilizing food, or even as a source of power.”

“Power?” Eddie said. “Boy, it must have been a strong isotope.” He knew that the strength of radioisotopes could be controlled largely by the length of time they were allowed to “cook” in an atomic reactor and soak up radioactivity.

“We weren’t planning to run a submarine with it,” his father said. “It wasn’t that strong. Still, it doesn’t take so very much radioactivity to make two ounces of an isotope quite powerful—and quite deadly. I only hope whoever stole it knows what he’s doing. However, I’m sure he does.”

33

“You mean he must have been an atomic scientist himself?” Eddie asked.

“Let’s just say he—or both of them—have enough training in the subject to know how to handle that isotope safely,” Mr. Taylor said.

“But, Dad,” Eddie wondered, “what could they do with it?”

“They could study it,” his father explained. “At least, they could send it somewhere to be broken down and studied. Being a new isotope, the formula is of great value.”

“What do you mean, send it somewhere?” Eddie asked.

“Perhaps to some other country.”

“Then—then you mean whoever stole it were spies!” Eddie exclaimed breathlessly.

“That’s entirely possible,” his father said. “In fact, it’s the only logical explanation I can think of. People simply don’t go around stealing radioactive isotopes without a mighty important reason.”

“Dinner’s ready,” Eddie’s mother called from the kitchen.

34

During dinner Eddie wasn’t sure just what he was eating. The idea of spies stealing atomic materials kept building up in his mind. By the time dessert was finished, he was anxious to talk with someone, yet he knew he shouldn’t bother his father with any more questions. He asked if he could go over and visit with Teena for a while.

“Well, you were together most of the day,” his mother said, “but I guess it’s all right. Be back in about an hour, though.”

It was a balmy evening. On such evenings, he and Teena sometimes walked along the beach barefoot, collecting sea shells. Today Eddie had no desire to do that. He ran down the block.

Teena answered his knock.

“Come on in, Eddie,” she invited, seeming surprised to see him. “Mother and I are just finishing dinner.”

“Oh, I figured you’d be through by now,” Eddie apologized, following her inside.

“Hello, Eddie,” Mrs. Ross said, but she didn’t seem as cheerful as usual.

35

“Good evening, Mrs. Ross,” Eddie said. “I—I hope I’m not making a pest of myself.” He looked around for Mr. Ross, but Teena’s father apparently hadn’t arrived home from Acme Aircraft yet. There wasn’t a place set for him at the table, either.

“You’re never a pest, Eddie,” Mrs. Ross assured him. “I was going to call your mother in a little while about that newspaper write-up.”

“Oh, you read it?” Eddie said.

“How could anyone miss it?” Teena said. “Right on the front page.”

“I suppose your father is quite concerned over it,” Teena’s mother said.

“Oh, yes,” Eddie affirmed. “He was the one who ordered the isotope.”

“What’s an isotope?” Teena asked.

“I’m not sure I know, either,” Mrs. Ross said. “Maybe we could understand more of what it’s all about if you could explain what a radioisotope is, Eddie.”

“Well,” Eddie said slowly, “it’s not easy to explain, but I’ll try. You know how rare uranium is. There’s not nearly enough of it to fill all the needs for radioactive materials. Besides, pure uranium is so powerful and expensive and dangerous to handle that it’s not a very good idea to try using it in its true form. So they build an atomic reactor like the one at Drake Ridge.”

36

“We’ve driven by it,” Mrs. Ross said. “My, it’s a big place.”

“I’ll say,” Eddie agreed. “Of course, only one building holds the reactor itself. It’s the biggest building near the center.”

“I remember it,” Teena said.

“Well, the reactor is about four stories high,” Eddie went on. “They call it a uranium ‘pile.’ It’s made up of hundreds and hundreds of graphite bricks. That’s where they get the name ‘pile’—from brick pile. Anyway, scattered around in between the bricks are small bits of uranium. Uranium atoms are

radioactive. That is, they keep splitting up and sending out rays.”

“Why do they do that?” Teena asked.

“It’s just the way nature made uranium, I guess,” Eddie said. “Most atoms stay in one piece, although they move around lickety-split all of the time. Uranium atoms not only move around, but they break apart. They shoot out little particles called neutrons. These neutrons hit other atoms and split them apart, sending out more neutrons. It’s a regular chain reaction.”

37

“I’ve heard of chain reactions,” Mrs. Ross said.

“Well, with all of the splitting up and moving around of the uranium atoms,” Eddie went on, “an awful lot of heat builds up. If they don’t control it—well, you’ve seen pictures of atomic-bomb explosions. That’s a chain reaction out of control.”

“Out of control is right,” Teena said.

“But the atomic piles control the reaction,” Eddie said. “The graphite bricks keep the splitting-up atoms apart so one neutron won’t go smashing into other atoms unless they want it to. They have ways of controlling it so that only as much radiation builds up as they want. You can even hear the reactor hum as the radioactive rays go tearing through it. But by careful tending, the scientists keep the atomic collisions far enough apart so the thing doesn’t blow up.”

38

“Boy, that sounds dangerous,” Teena said.

“Well, they know just how to do it,” Eddie replied.

“Aren’t the rays dangerous?” Mrs. Ross asked.

“I’ll say they’re dangerous,” Eddie said. “But the whole pile is covered by a shield of concrete about eight feet thick. That keeps the rays from getting out and injuring the workmen.”

“Goodness. Eight feet is a lot of cement.”

“It takes a lot to stop radioactive atomic particles,” Eddie explained. “Especially the gamma rays. They’re the fastest and most dangerous, and the hardest to stop. Alpha and beta rays are fairly easy to stop. But the gamma rays are regular high-velocity invisible bullets. They’ll go right through a stone wall unless it’s plenty thick. Of course, you can’t see them. Not with even the most powerful microscope in the world.”

“I wouldn’t want to work around a place where I might get shot at by—by dangerous rays you can’t even see,” Teena said.

39

“I would,” Eddie said. “Everyone is carefully protected. They see to that. Well, anyway, if all of those uranium atoms were shooting radioactive rays around inside of that pile and doing nothing, there would be an awful lot of energy going to waste. So the atomic scientists take certain elements which aren’t radioactive, but can be made radioactive, and shove small pieces of them into holes drilled in the pile.”

“Isn’t that dangerous?” Teena asked.

“They don’t shove them in with their bare hands,” Eddie said, trying not to show exasperation. “They use long holders to push the small chunks of material into the holes in the reactor. Then, as those uranium atoms keep splitting up and shooting particles around inside of the pile, some of them smack into the chunks of material, and stick there. Most elements will soak up radiation, just like a sponge soaks up water.”

“My, that’s interesting, Eddie,” Mrs. Ross said.

“I’ve seen them do it,” Eddie said proudly, then added, “from behind a protective shield, of course. When the material has soaked up enough radiation, they pull it back out. They say it’s ‘cooked.’”

“You mean it’s hot?” Teena asked.

“It’s hot,” Eddie said, “but not like if it came out of a stove. By hot, they mean it’s radioactive. If you touched it, or even got near it, you would get burned, but you probably wouldn’t even know it for a while. It would be a radiation burn. That’s a kind of burn you don’t feel, but it destroys your blood cells and tissues, and—well, you’ve had it.”

“So that’s what a radioisotope is,” Mrs. Ross said. “It’s like a sponge. Only instead of soaking up water, it soaks up radiation.”

“That’s about it,” Eddie said. “My dad says that as more is learned about the ways to use isotopes, the whole world is going to be improved. You’ve heard of radiocobalt for curing cancer. Well, that’s an isotope. They make it by cooking cobalt in an atomic reactor. Oh, there are hundreds of different isotopes. Like I said, isotopes can be made of most of the elements. And there are over a hundred elements. Some soak up a lot of radioactivity, and are strong and dangerous. Others absorb only a little and are pretty safe to use. Depends, too, on how long they let them cook in the reactor.”

“What kind was the one stolen from the college today?” Teena asked.

“Dad didn’t say exactly,” Eddie answered, “except he did say that if whoever took it didn’t know what he was doing and opened up the lead capsule, it could kill him. Of course, even the mild isotopes are deadly if they’re not handled right.”

“My goodness, it is a serious matter, isn’t it?” Mrs. Ross said.

Eddie nodded. It was even more serious than its threat of danger to anyone who handled it carelessly. It was a new isotope—a secret isotope. His father hadn’t said whether it had been developed for curing things or for destroying things. But many radioisotopes could do either; it depended on how they were used. Eddie assumed that anyone who would stoop to stealing isotopes more than likely would be interested in their ability to destroy rather than their ability to benefit mankind.

“Well, I certainly do hope everything works out all right,” Teena’s mother said.

“So do I,” Teena agreed.

Eddie glanced at the kitchen clock. “Oh, boy,” he said, “I’d better be heading back home. I didn’t mean to come over here and talk so long.”

“Oh, we’re glad you did, Eddie,” Mrs. Ross said. “I’m afraid too few of us know anything about this atom business.”

“That’s right, Mrs. Ross,” Eddie agreed. “People should talk more and read more about it. After all, this is an atomic age. We might as well face it. My father says that in horse-and-buggy days everyone knew how to feed a horse and grease a wagon wheel. They knew what was needed to get the work done. But now that atoms are being harnessed to do the work, not many people even bother to find out what an atom is.”

Mrs. Ross smiled. “I guess you’re right, Eddie,” she said, “but I wouldn’t quite know how to go about feeding an atom.”

“Or greasing one,” Teena added.

Eddie laughed. “I sure wouldn’t want the job of trying to feed a herd of them the size of a period,” he said. “Did you know that there are about three million billion atoms of carbon in a

single period printed at the end of a sentence. That's how small atoms are."

"Three million billion is a lot of something," a man's voice spoke behind him. "What are we talking about, Eddie?"

"Oh, hello, Mr. Ross," Eddie said, turning around and standing up. "I didn't hear you come in."

Teena's father was a medium-sized man with light-brown hair which was getting somewhat thin on top. He was usually quite cheerful and full of fun, but tonight his face seemed unusually drawn and sober. He stepped to the table, leaned over, and gave both Teena and Mrs. Ross a kiss on the cheek.

44

"Eddie was telling us about atoms," Teena's mother said. "Did you know there were three million billion of them in a period?"

"How many in a comma?" Mr. Ross said to Eddie, then added quickly, "forget it, Eddie. It wasn't very funny. I—I'm afraid I don't feel very funny tonight."

"Sit down, dear," Mrs. Ross said. "I'll warm your dinner. You didn't sound very cheerful when you called to say you would be late. How did everything go at the plant today?"

"Not so good," Teena's father said tiredly. "In fact, not good at all."

Problems. It seemed that everyone had problems, Eddie thought, as he started to leave.

45



CHAPTER THREE

Three days later Eddie learned the nature of the trouble at Acme Aircraft Company. It was midmorning Saturday. Carrying a picnic lunch, he and Teena were hiking along the beach toward the distant U.S. Coast Guard lighthouse which stood on a high point overlooking Moon Bay. Old Captain Daniels, the lightkeeper, had been a friend of theirs for nearly two years. Every once in a while Teena and Eddie went to visit “Cap,” as they fondly called him. Teena would put up a picnic lunch which they shared with the kindly old man whose lonely vigil over the light had warned ships off the rocky coast for a good many years.

Eddie wasn’t sure exactly how the conversation got around to Acme Aircraft Company. It seemed that Teena mentioned something about trouble. Eddie asked, “What trouble?”

46

“Didn’t you read the paper this morning?” Teena asked.

“We don’t take the morning paper,” Eddie said. “We take the *Globe*.”

“The aircraft company has been keeping it quiet,” Teena said, “but somehow the news leaked out. It was all in this morning’s paper.”

Actually, Eddie doubted that he would have noticed it, even if they had taken the morning paper. The big thing around his house was the disappearance of the radioisotope. No promising clues had been found. The theft of the isotope remained as much a mystery as it had been from the first day.

The few times he had seen his father, he had noticed how much deeper the lines of worry on his face had become. Eddie had avoided asking questions, yet he couldn't help wondering if his father was in danger of losing his job at the college. Some of the things his mother had said seemed to hint at the possibility.

The newspapers had taken a dim view of the robbery. They protested against the carelessness which would enable the theft, in broad daylight, of such a valuable, secret, and hazardous thing as a radioactive isotope.

47

The blame, of course, fell primarily upon Mr. Taylor's shoulders.

"Eddie," Teena said, "you're not even listening to me."

"I—I'm sorry," Eddie answered. "What did you say?"

"I merely said that some important blueprints are missing from my father's department at Acme Aircraft," Teena explained.

"Maybe someone put them in the wrong drawer or something," Eddie suggested.

"They've looked everywhere, Eddie," Teena said. "That's why my father's been getting home late every day. They've searched absolutely everywhere."

"Well, I guess they can always make new blueprints," Eddie said. He really couldn't see why it was so important, especially not if you compared it to stolen radioisotopes.

48

"That's not the point," Teena said sharply. "They were top-secret blueprints—something to do with guiding a new missile Acme Aircraft is getting ready to make. If the plans were stolen—well, you know what that could mean."

The importance of what Teena was saying struck Eddie suddenly. Could there be any connection between the missing blueprints and the stolen isotope?

The idea sent a chill along Eddie's spine. Perhaps there was a whole spy ring operating around Oceanview!

He mentioned it to Teena.

"I thought the same thing," she said. "But, Eddie, we're not at war or anything. It's silly to think there are spies and things like that running around. That's comic-book stuff."

"Not to me, it isn't," Eddie said. "There's plenty of spying going on, war or no war. Every once in a while you hear about it."

"Aw, you're just trying to scare me," Teena said. She stopped and picked up a shell, looked it over, then skipped it into the surf. Sandy chased it, yipping happily, but turned and scurried back just ahead of the foaming surf.

49

Eddie smiled. "Maybe I am," he said. "Maybe I'm scaring myself, too. Anyway, if we're going to hike all the way to the lighthouse, we'll have to hurry up."

As they walked on up the coast, the sandy beach gave way to rock formations which jutted out into the ocean. They picked their way carefully over the rocks. Now and then they stopped to inspect some tide pool for small crabs and other sea life left by the receding water. Sandy was beside himself with joy as he chased small crabs into rock crevices.

Teena found a starfish which she dropped into a small cloth sack she had brought along. Eddie had never been very interested in gathering shells and other sea souvenirs, but Teena had quite a collection at home.

They crossed over the rocks and dropped down into a sandy cove.

“There’s someone with a boat,” Teena said, pointing along the curving beach. A rowboat was pulled up on the sand. Two men stood beside it.

50

“Fishermen,” Eddie said. “Let’s see if they caught anything.”

There were several other small boats out on Moon Bay. Eddie and his father had fished the bay several times themselves. Although shallow in places, there were spots in the bay where good-sized perch and bass, and occasional halibut were caught.

Eddie and Teena hurried along the beach. The two men looked up as they approached.

“Hi, there,” Eddie greeted. “Any luck?”

The two men glanced at each other, seeming to pass a silent question back and forth.

Eddie laughed. “It’s O.K.,” he said. “I know most of the fishing spots out there. You won’t be giving away any secrets.”

51



“Hi, there,” Eddie greeted. “Any luck?”

“Secrets?” one of the men said. He was tall and thin. His cheekbones pushed sharply outward against the sides of his face. His skin was strangely white for that of a fisherman in midsummer. Most week-end fishermen around Oceanview had pretty good tans by this time. Both men wore faded blue denims, white sneakers, and bright-colored sports shirts. The fact that their clothes looked new made Eddie think it might be the first time they had fished Moon Bay. The orange-and-white rowboat pulled up on the sand had been rented from Anderson’s Landing. Both Eddie and Teena knew Mr. Anderson well.

“We know fishermen don’t like to give away their fishing secrets,” Teena said, “so if you caught any, you don’t need to tell us where you got them.”

The other man smiled then. He looked relieved, Eddie thought. In appearance, he was almost the opposite of his companion. He was short and squat, almost fat. Despite the slight cool breeze from the ocean, the warm sun made his chubby face glisten with sweat. He seemed a little more willing to smile than the tall man. Eddie didn’t feel uncomfortable under his gaze, as he did under the stare of the tall man.

“To tell you the truth,” the short man said, “we haven’t been fishing yet. So I guess you couldn’t say we’ve had any luck.”

53

“Oh, I see,” Eddie said thoughtfully. Sandy began sniffing around the rowboat.

“Get that mutt away from there,” the tall man said.

“He won’t hurt anything, mister,” Eddie assured him.

He went over, though, and took hold of Sandy’s collar. As he did so he glanced into the beached rowboat. There were no fish, or even signs of fish. There were a couple bamboo poles which Eddie recognized also as having been rented from Anderson’s Landing. There was a box, probably the men’s lunch.

And under the plank seat stretching across the beam Eddie saw a round metal cylinder. At first he thought it was the kind of tube used as a carrying case to hold the sections of a jointed trout rod, but as he got a better look, it didn’t seem long enough for that.

Besides, who would use a light trout rod for ocean fishing? It wouldn’t be any good to catch the big bass which were

54

sometimes caught in the bay. It'd probably snap in two if you tried to horse a halibut in with it.

"What are you looking at, kid?" The tall man's harsh voice jerked Eddie out of his thoughts.

"N-nothin'," Eddie said.

"Then stay away from the boat."

"Take it easy, Simms," the short man said. "These kids don't mean any harm. They—they're not trying to steal our fishing secrets. Now, are you?" He smiled at Teena, displaying a mouthful of yellowish uneven teeth.

Looking at those teeth made Eddie mighty glad his teeth would never look like that. What little bother his braces and the cleaning were would sure be worth it in the long run. He never wanted yellow, uneven teeth like that man had.

"I should say we wouldn't try to steal any fishing secrets," Teena answered the fat man's question. "You're welcome to all the fish you can catch."

"We don't care how you catch them, or what with," Eddie added, "long as it's legal."

55

"Anyway, we're on our way to visit Captain Daniels at the lighthouse," Teena said. "Come on, Eddie."

"Don't go away mad," the heavy-set man said. "We didn't mean any harm."

"Let 'em go, Roy," Simms said. "We've got work to do."

Eddie motioned to Teena and called Sandy. He had intended to tell the men of a good fishing spot only a few hundred yards out from the cove, but the way the men acted made him change his mind.

At the far end of the cove, Eddie and Teena stopped and turned to watch the two men as they shoved the rowboat into the calm surf and climbed in clumsily over the side.

“Boy, I’m glad all fishermen aren’t like that,” Teena said. “That tall man sure acted mean. I hope they don’t catch any fish.”

“I don’t think they will,” Eddie said. “I saw their bait can. Know what they’re using?”

“Sand crabs?”

“No. That’s what they should be using. They had some old dried up mussels. The fish here in Moon Bay don’t bite on mussel. Dad and I have tried it.”

56

“Then I wonder why Mr. Anderson sold it to them,” Teena said. “Mr. Anderson usually helps the fishermen. It’s good for his boat-rental business to sell the right bait.”

“I’ll bet they didn’t even ask what kind of bait was best,” Eddie said. “They probably grabbed the first thing they came to. And Mr. Anderson always has a few mussels in his bait bins.”

“I didn’t think fishing was so good in the middle of the day,” Teena said. She pointed out across the water. “See, most of the boats have gone ashore.”

“That’s right,” Eddie said. “If those guys wanted to catch fish they should have been out there early this morning when the big ones were biting.”

“Guess they don’t know much about fishing, huh, Eddie?” Teena said, smiling.

57

“That’s what I figure,” Eddie agreed. “Besides, they didn’t even act like fishermen. That tall fellow really was a grouch. First time I ever ran across a grouchy fisherman.”

“Anyway, let’s quit worrying about them,” Teena suggested. “It’s almost noon. We want to reach the lighthouse before Cap has lunch. He can’t very well eat his lunch and ours, too.”

“O.K.,” Eddie agreed, taking one last glance at the two men rowing out on the blue water of the bay. “But something smells fishy about those two—and I don’t mean the kind you catch on a hook!”



CHAPTER FOUR

The lighthouse was a tall concrete finger, painted dazzling white with broad red rings around it. It stood on the top of a rock palisade which rose steeply from the beach. Steel stairs spiraled upward on the outside, leading to the strong glass-enclosed electric eye at the top.

Eddie and Teena paused on the beach below and looked up. Crude steps hewn out of the rocks led up to the lighthouse.

“I’ll carry the lunch,” Eddie volunteered. “And be careful. The sea spray can make those steps slippery.”

They took their time getting to the top. Sandy went ahead, sniffing in every crevice on the way.

59

“Phew!” Teena gasped as they reached the base of the lighthouse. “There seem to be more steps every time we climb it.”

Eddie smiled and shifted the lunch sack to his other hand. “You’re getting old, Teena,” he teased.

“Welcome aboard, mates,” a deep, kindly voice spoke from nearby.

They turned and saw Captain Daniels standing outside the door of his living quarters, a tiny three-room cottage located about fifty feet from the base of the lighthouse.

“Oh, hello, Captain Daniels,” Teena called. “Sure glad you’re home.”

“Home?” the former sea captain said, smiling. “A lightkeeper is always home.”

In Eddie’s opinion, Captain Daniels looked exactly like an old ship’s captain or a lightkeeper should look. He wore a fringe of white beard which formed a half-circle, starting under one ear and curving across his chin and up the other side. His bushy white hair fairly exploded from beneath the battered dark-blue seaman’s cap which he wore even while eating. Eddie sometimes wondered if Captain Daniels wore the cap to bed.

60

The old mariner also had sharp blue eyes. Eddie pictured all stout seamen as having sharp blue eyes.

“We brought a little lunch with us, Captain Daniels,” Teena said. “Hope you haven’t eaten already.”

A twinkle came into Cap’s eyes. “I might have,” he said, “but I reckon I better confess that I saw you through my telescope coming up the beach. Thought I’d better hold off on lunch—just in case.”

“Can we eat outside?” Teena asked.

“The lawn’s nice and dry,” Cap said.

“Let’s make it a picnic,” Eddie suggested.

“Good idea, mate,” the retired seafarer said.

Captain Daniels took great pride in his small patch of grass. It seemed to grow right out of the rock on which the lighthouse stood. However, Captain Daniels had hauled in topsoil from miles away and spread it carefully to make the lawn. He

61

tended it, and the flower beds which bordered it, with an affection that seemed strangely out of place for a swashbuckling ship's captain who had roamed the seven seas.

The three of them sat down on the lawn. Teena passed around the sandwiches, opened the potato chips, and unwrapped the pickles and olives.

They ate for a while in silence, looking off across the blue water of the bay toward the open ocean beyond. Eddie's gaze followed the curving shore line to the north. Land's end in that direction was Cedar Point, which stuck its rocky finger out into the ocean. It was wildly rugged country, difficult to get to except by boat across the bay. Eddie supposed that was why the lighthouse had been built on the smaller point located on the more civilized curve of the bay. Yet the lighthouse was high and plainly visible to ships at sea.

Captain Daniels finished his lunch, dug a pipe from his pocket, and tamped tobacco into the bowl. "Mighty good," he said. "Sure nice of you young folks to share your rations with me."

62

"Oh, we like to do it, Captain Daniels," Teena said. "It's so much fun coming up here to visit you."

"From what I've been reading in the papers," the lightkeeper said, "I hardly expected to see you for a while."

"You mean the stolen isotope?" Eddie asked.

"I don't know much about isotopes," Cap said, "but I do know that the newspapers have been making your father walk the plank for letting it be stolen."

"It really wasn't his fault," Eddie defended.

“Of course not,” Captain Daniels agreed. “But someone always gets blamed. Just like those missing blueprints I read about in this morning’s paper. Teena’s father probably has nothing to do with guarding them, but when they turn up missing, he’s the one who gets lashed to the mast. The captain of a ship takes the blame for everything that happens aboard. Actually, that’s the way it should be, I suppose.”

Eddie had to agree, but he didn’t like to think about the worry his father and Mr. Ross were going through. He had been trying not to think about it.

63

Captain Daniels seemed to sense this. He quickly changed the subject.

“Don’t seem to be many fishermen out today,” he said, looking off across the bay. “And there’s one boat out there that could just as well have stayed ashore. Won’t catch anything worth frying out there on top of the sand bar.”

The rowboat had been anchored over the light-blue strip of water which marked the familiar sand bar stretching nearly a half mile across the middle of the bay. The sand bar lay about ten feet beneath the surface of the water. It was marked by three buoys, one at each end and one in the middle. Deep-draft boats avoided the sand bar. Fishermen kept away from it, as the larger fish lay in deeper water.

“Isn’t that the boat with those two men, Eddie?” Teena asked.

64

“I think so,” Eddie said, squinting through the sunlight.

“What two men?” Cap asked.

Quickly Eddie told him about the two strangers he and Teena had come across at the cove. Captain Daniels reached into his pocket and brought out a small telescope. He pulled its

sections out to full length and handed it to Eddie. “See for yourself,” he invited.

Eddie adjusted the lens to his vision. With the telescope it was easy to see that the two men in the rowboat were the tall one called Simms and the chunky one called Roy.

“Anyway,” Eddie said, “they don’t seem to be pulling in any fish.” He passed the telescope to Teena.

“It doesn’t look like they’re even trying,” Teena said. “There’s only one line in.”

“Maybe they’re just relaxing,” Captain Daniels said. “Some people don’t care whether they catch any fish or not. They rent a boat, row it out and anchor it, and then sit around soothing their nerves. People build up a lot of tensions these days, you know. Folks have different ways of getting rid of them.”

65

“They were nervous, all right,” Teena said. “Especially the tall one.” She handed the telescope back to Captain Daniels.

“Well, let’s forget about them,” Eddie suggested. “Captain Daniels, would you like us to help polish the light again today?”

“You know you’re always welcome to help with that,” the lightkeeper said, “but I don’t want you coming up here thinking I expect you to work.”

“Oh, but that isn’t work,” Teena said. “It’s fun.”

Eddie agreed with that. Not only was it fun, but it was a great thrill to climb up to the top of the lighthouse.

Captain Daniels got some rags and a can of window cleaner out of a small tool shed at the foot of the lighthouse.

“Why don’t you let us do it today, Captain Daniels?” Teena asked. “No use in your climbing all of those stairs.”

66

“You win,” the lightkeeper said, smiling. “I’ll wait down here.”

Eddie and Teena took the rags and cleaner and started up the steel stairs which spiraled up the outside to the top of the lighthouse. The stairs were perfectly safe, as a waist-high railing prevented any possibility of an accident.

Reaching the top, they paused on the narrow steel balcony that circled the light. The view across the bay was spectacular—blue water and whitecaps as far as they could see. A couple of steamers dragged banners of smoke across the distant horizon. In the other direction they saw Oceanview sprawling out inland from the shore of the bay. Both Acme Aircraft Company and the college campus were in plain view.

After filling themselves with the view, they got busy on the light. It was like polishing a giant lantern chimney. It had thick, wavy glass to magnify the beam of the enormous electric lamp which rotated inside, making three complete turns a minute. Being daytime, the light was turned off. In fact, Eddie never had seen the light up close at night. He imagined it would be very blinding, although he doubted if anyone ever would be foolish enough to climb up and look into it. It was bright enough, even from a distance, as it swept its white warning finger through the sky.

67

He and Teena worked away at spreading the window cleaner. After it had dried on the thick glass, they went over it carefully with their soft rags. The dirt and the white deposit left from the salt spray came off easily, leaving the glass bright as crystal.

“I guess that’s it,” Eddie said, after they had made a complete circle of the glass. He paused to take one last look around.

“We’d better be getting back home, too,” Teena suggested. “It must be three o’clock.”

Eddie glanced up at the sun. “You’re about right,” he said.

They made their way back down the stairs. Cap was waiting at the bottom.

68

“It’s as bright as the northern star, mates,” he said, craning his neck to get a good look at their handiwork. “I sure do thank you both.”

“We’re the ones to thank you for letting us come out here to visit you, Captain Daniels,” Eddie said.

“Any time,” the old mariner invited. “You’re always welcome. And I don’t expect you to bring a lunch or polish the light, either.”

“We have to go now,” Teena said. “But we’ll come out to see you again before long. Come on, Sandy.”

“I’ll be looking for you,” Captain Daniels called after them, as they started down the rock steps toward the beach.

Later, when they reached the cove they noticed that the rowboat was no longer anchored out over the sand bar. Then Eddie saw it in close to shore, heading for Anderson’s Landing. He didn’t give it any more thought.

As they approached Anderson’s Landing, the two strangers were tying up at the dock.

69

“Let’s see if they caught anything,” Eddie suggested.

“Let’s not,” Teena objected. “They weren’t very nice to us.”

“They didn’t mean anything,” Eddie said. “Maybe someone should tell them that the fishing is no good over the sand bar.”

“I’ll bet they found that out for themselves,” Teena said.

But Eddie already had started walking out onto the plank boat dock. Teena followed.

“Here, mister, I’ll help you,” Eddie offered as the heavy-set man removed the oars from the oarlocks and moved toward the prow of the boat.

“O.K.,” the man said, trying to keep his balance in the rocking boat. Then he glanced up. “Hey, you’re the kids we saw earlier, aren’t you? You following us or something?”

“No, sir,” Eddie said. “We were on our way home. Just thought we’d come out and see what kind of luck you had.”

“We did all right, didn’t we, Roy?” the tall man said.

70

“But where are your fish?” Teena asked.

“We left them in the bay,” Roy, the portly man, said.

“I guess so,” Eddie said, smiling. “No one ever catches any fish out over the sand bar. The fish hang around in the deeper water.”

“Well, we don’t care much for fish, anyway,” Roy said.

“Then why do you go fishing?” Teena wondered.

“We do it to get away from kids who ask silly questions,” Simms said curtly. “Now beat it and leave us alone.” He tossed the two fishing poles onto the dock and climbed out of the boat.

“Sure, mister,” Eddie said. “We didn’t mean to bother you.”

“Don’t get sore, kids,” Roy said. “Simms is a little sunburned, that’s all. Makes him cranky.”

The tall man was sunburned, all right. Eddie had noticed that. But then, he had expected it. Neither man boasted any kind of a tan, and the sun had been hot all afternoon.

Eddie also had noticed something else. It struck him as strange, although he didn't know what to make of it. The metal tube which he had noticed in the bottom of the boat when they had first met the men in the cove was no longer in sight.

If it had contained a collapsible fishing rod as he had guessed, why wasn't it still there in the bottom of the boat? Eddie was certain the men hadn't put in to shore between the time they had left the cove and now. If they had he and Teena would have noticed it from the lighthouse.

A metal tube like the one Eddie had seen earlier in the bottom of the rowboat simply would not disappear. Perhaps it hadn't contained a collapsible fishing rod, as he had guessed. If not, what was in the cylinder?

And where was it now?



CHAPTER FIVE

Nearly a week went by. The lead capsule containing the stolen radioisotope had not been found. In fact, as far as Eddie knew, there had been no worthwhile clues on which to base a search. Curious as he was, Eddie still managed to keep from asking his father a lot of questions. Around home, Mr. Taylor had been thoughtfully silent. Eddie knew that his father must be very worried.

Eddie gathered enough from the conversations between his parents to know that the search for the stolen isotope was still going on. In fact, it was pretty well known that FBI agents had arrived in Oceanview to lend a hand. Eddie hadn't seen them, but several of his friends had. It was hard to keep secrets in a college town like Oceanview.

73

The newspapers had temporarily dropped the story. After all, when the radioisotope had been stolen it had made a big story, but nothing more had happened, so there was nothing more to write about.

There were still articles in the newspapers about the Acme Aircraft Company problem of the missing blueprints. Actually, it seemed to Eddie that the newspapers were making more of a mystery out of the missing blueprints than of the stolen radioisotopes. Perhaps that was because it had not yet been decided whether the blueprints had been stolen, destroyed by accident, or simply lost. In an aircraft plant, where thousands and thousands of plans are being used at all times, some carelessness is apt to occur. Eddie found it hard

to believe that anyone could get careless with top-secret blueprints, yet such things did happen.

Maybe the reason people remained curious about the missing blueprints was that everyone knew what a blueprint was. Even the word “radioisotope” meant very little to most readers. What they were and what they did were even less well known.

74

Friday morning Teena came whistling up to the back door at Eddie’s house to see if he wanted to do anything.

“Mom’s gone shopping,” Eddie said. “She’ll be back in a few minutes. Then maybe we could take the Geiger counter and —”

His words were interrupted by the telephone ringing inside.

“Be right back,” he said, hurrying into the house.

His father was on the other end of the line.

“Eddie,” he said, “there’s a dark-green notebook on my desk in the study. I forgot it this morning. Can’t get away from here, and I need it.”

“I’ll bring it over, Dad,” Eddie volunteered quickly.

“Good. I’ll be outside the botany building. Know where that is?”

75

“Botany? Where they raise all the plants and stuff?” Eddie asked.

“That’s right.”

“I know where it is,” Eddie said. “Be there in five minutes.”

He went into the study, got the notebook off the desk, and went back outside.

“I’ve got to tear over to school with this notebook,” he explained to Teena. “That was my dad.”

“I’ll go with you,” Teena said.

“O.K.”

It was only a few blocks to the college campus. Reaching the grounds, they took a short cut past the men’s gymnasium, crossed the athletic field, and arrived at the Botany Building.

“I see Dad over there,” Eddie said, pointing. There were several men standing in a group in the small cultivated field which the botany department used to grow test plants of various kinds. Eddie and Teena picked their way carefully between the rows.

76

“You made good time, Eddie,” his father said, taking the notebook. “Morning, Teena. Hope I didn’t interrupt any big plans.”

“Oh, no, Mr. Taylor,” Teena said. “We weren’t doing anything special.”

Eddie was about to turn and leave, when he noticed that several of the young men—students, no doubt—were wearing strange-looking, long, heavy gloves.

“Why the gloves, Dad?” he asked.

Mr. Taylor glanced up from the notebook. “We’re experimenting with radioactive tracers,” he said. “They’re weak—certainly not dangerous—but there’s no harm in taking a little extra precaution. The gloves are lead-lined and absorb any of the rays which might possibly be picked up from handling the plants.”

“What are tracers, Mr. Taylor?” Teena asked.

Eddie shot her a quick scowl. He doubted that his father wanted to be bothered with questions at the moment. Then Eddie noticed one of the students holding the wandlike probe—or diode—of a Geiger counter down close to the leaves of one plant. It seemed a strange thing to be doing. Who would prospect for uranium in plants.

“That’s a good question, Teena,” Mr. Taylor said. “Glad to see you’re interested.” He turned and spoke to the student with the Geiger counter. “Jim, you want to explain to this young lady, and my son here, what tracers are?”

Eddie smiled to himself. His father was the kind of a professor who believed his students should have the chance to use their knowledge whenever possible.

“Maybe we should all listen to this,” Eddie’s father said to the others.

They gathered around as the dark-haired student switched off the Geiger counter, swallowed a bit uncomfortably, and turned to face the outdoor classroom which Teena’s question had formed on the spot.

“Well, er—” Jim began, addressing Teena, “a tracer is a radioactive isotope which we—well, send out on a journey, then follow it with a Geiger counter.”

“I’m afraid that’s not too clear, Jim,” Mr. Taylor said. “Teena’s not an atomic scientist. Can you simplify it for her?”

“Oh, I know what a radioactive isotope is, Mr. Taylor,” Teena said proudly.

“You do?” Eddie’s father said in surprise.

“It’s something—I think you call it an element—which they put in an atomic reactor, and—and cook it until it becomes

radioactive. Kind of like a sponge soaking up water.”

“Very good,” Mr. Taylor nodded, obviously impressed.

“Eddie explained it to me,” Teena said, smiling, “but he didn’t say anything about tracers.” She turned her attention back to Jim, the student.

“All right,” Jim said, seeming more at ease, “let’s look at it this way. Any radioisotope keeps shooting out rays. Of course, you can’t see the rays with your eyes. They’re almost too small to think about. But you can follow them with a Geiger counter.” He indicated the black metal instrument which he still held in his hand.

79

“Well,” Jim went on, “say, for instance, that you wanted to know how fast a stream of water flows. You might toss in a rubber ball and time how long it takes to float a mile downstream. That would give you its speed. Or say you wanted to know which way its currents twist and turn. You might dump in a gallon of ink and watch it follow the currents. In a way, the ball and the ink are tracers. Not radioactive tracers, of course, but by watching how they act, you learn what you want to know.”

“Let’s talk about radioactive tracers,” Eddie said eagerly.

“You’re crowding me, bub,” Jim said, smiling. Everyone laughed.

“How are we using tracers here, Jim?” Mr. Taylor prompted.

“We’re testing the use of phosphate in plant growth,” the student explained. “We want to know what the plant does with it. Does a phosphate fertilizer merely feed the plant’s roots, or is it pulled up into the stems and leaves? And we want to know how quickly the plant absorbs it, if at all. Of course, we can’t see it, but if we make the phosphate slightly radioactive,

80

then we have what we call a tracer. By using a Geiger counter, we can follow or trace its movement.”

“Can you explain our method, Jim?” Mr. Taylor said.

“Well, we spread a little of the radioactive phosphate around the plant,” the student said. “Soon the roots start taking it in.”

“How do you know that, Jim?” Mr. Taylor asked.

“We hold the Geiger counter to the root. If it starts clicking faster than usual, we know the root has absorbed some of the phosphate tracer. We also hold the Geiger counter over the stems and leaves. As the tracer works upward into the plant, the Geiger counter reacts to it. Here, let me show you how it works on this cotton plant.”

Eddie and Teena moved over closer to the two-foot-high plant. Jim switched on the Geiger counter. Eddie saw the needle on the gauge flutter slightly, indicating the normal cosmic-ray background count.

81

“Teena,” Jim said, handing her the earphones which were attached by a long wire to the Geiger counter, “you take these earphones. Now, I’ll pass the probe down close to the base of the cotton bush.”

“What does the stick do?” Teena asked. Although Eddie had explained it to her, she seemed to feel that, as a pupil, she should ask some questions to help Jim out.

“Stick? Oh, you mean the probe. Actually, it’s called a diode, but probe’s easier to remember. Anyway, the probe is a vacuum tube filled with a special kind of gas. Whenever invisible radioactive particles shoot through the probe and into the gas, the Geiger counter clicks, and the needle on the dial moves forward. The more rays shooting through the probe, the more clicks; the more clicks, the more

radioactivity. That's why Geiger counters are so useful in hunting for uranium. Uranium is very, very radioactive. If you happen onto some uranium ore, the Geiger counter really goes wild."

"We have a Geiger counter at home," Eddie said eagerly. "Teena and I have gone uranium prospecting several times."

82

"Haven't found any uranium," Teena said, "but we've had fun trying. Whoops. There's some clicking!" She put her hands up to the earphones.

Jim had moved the probe down close to the stem of the cotton plant.

"Good," he said. "We mixed a little radioactive phosphate into the ground around the roots this morning. See, the Geiger counter shows that the phosphate tracer has already started moving up into the plant. Helps show how important phosphate is to plant growth, and how eagerly the plants absorb it."

"The plant sure looks healthy enough, all right," Eddie said.

"Right," Jim said. "Now let's see how far up into the plant the tracer has gone."

He moved the probe upward over the smaller twigs and leaves. On the lower leaves the Geiger counter kept clicking rapidly. Eddie watched the needle stay forward on the gauge.

83

"See, the leaves have taken a lot of it in already," Jim explained.

Then, as he moved the probe farther up toward the top of the plant, the clicking diminished until only the familiar slow background count remained.

“It quit,” Teena said.

“Shows that the phosphate has only reached about half of the plant so far,” Jim said. “You see, with the tracer and the Geiger counter we can tell just how far it has gone and how long it has taken. We can even tell how much has been absorbed by comparing the amount of radioactivity in the leaves and stems of the plant to what we know was contained in the original tracer.”

“Boy, that’s something!” Eddie exclaimed.

“By adding tracers to some fertilizers,” Jim went on, “we found that the plant made no use of the fertilizer. The Geiger counter didn’t pick up any radioactivity in the plant. Meant wasted money to any farmer or gardener who used it. Now do you see what we mean by a tracer? See how radioactive tracers can be helpful?”

84

“Oh, yes,” Teena said. “I do.”

“I’ll bet if I had some of that tracer I wouldn’t lose so many things,” Eddie said. “I could paint a little on my marbles or sling-shot. Then I could always find them with a Geiger counter.”

“You could, at that,” his father said. “And I wouldn’t be stepping on the marbles in my bare feet. But, of course, great care must be taken in handling radioisotopes, which is what tracers are.”

Jim had warmed up to the subject, and wasn’t quite ready to drop it. “Tracers are used in many ways,” he went on. “They are used in medicine to locate diseased tissue which attracts and absorbs certain isotopes. A radiation-sensitive instrument, similar to a super Geiger counter, sniffs out the isotope and locates the damaged tissue. Then the doctor knows what to treat, or where to operate. Radioisotopes are used in various

85

food tests. By watching the tracer with electronic gadgets, they can tell whether the food is a muscle builder, a bone builder, or what.”

“You can make machine parts radioactive,” Mr. Taylor said. “Then by seeing how many radioactive particles are in the oil after the machine has been run, you can tell how much wear the machine has taken. Oh, there are hundreds of ways to use radioactive tracers. You might call them atomic signposts. Using a Geiger counter to read the signs, you are directed along the paths that lead to the answers of nature’s mysteries.”

“Wow!” Eddie exclaimed.

“Pretty flowery, at that, I guess,” his father said, smiling. “Well, anyway, Jim, you did a nice job of explaining it. Now, I think we’d better get back to our work. Thanks for bringing the notebook over, Eddie—and Teena.”

The two young people turned and started back toward Eddie’s house.

“Let’s go across the mall,” Teena suggested. “I haven’t been over here for a long time.”

The mall, as it was always called, was a broad ribbon of lawn which stretched for more than a block down the center of the college campus. It was bordered on both sides by the many buildings which made up Oceanview College. Sidewalks laced back and forth across the mall. During class changes, the area swarmed with students. Now, as Eddie and Teena walked along the mall, only a few students sauntered around or sat loafing in front of the buildings waiting for their next class.

Teena and Eddie walked past the library, the assembly hall, and the nuclear-science building. They were starting past the chemistry building, when Eddie tugged at Teena’s sleeve.

“Look,” he said, pointing to the back of a man walking about fifty yards ahead of them. “Who’s that?”

“Who’s who?” Teena asked. “You mean that man? Am I supposed to know him?”

“I think I do,” Eddie said. “He sure looks familiar.”

87

“How can you tell? All we can see is his back. Lots of backs look alike. He’s tall. Maybe he’s a basketball player. He looks older than most students, though. Why, his hair’s even a little gray, and—”

“I’ve got it now,” Eddie interrupted. “The tall and kind of gray part. You know who? Simms. That fellow we ran into down at the cove last week.”

“Well-ll, maybe,” Teena admitted thoughtfully. “We could tell for sure if he’d turn around. Anyway, I don’t see what difference it makes. Maybe he’s a student here. There are a lot of older students. Maybe he’s even a teacher. Lots of teachers fish on week ends. No reason to get excited.”

“Who’s excited?” Eddie challenged. “Can’t a fellow ask—”

“O.K., O.K.,” Teena said. “Anyway, there he goes into the chemistry building, so we’ll never know just who it was. And that’s the end of your mystery.”

Eddie didn’t say anything. He walked along, busy with his own thoughts. Probably Teena was right. Why make a mystery of it? Even if the man were Simms, what difference would it make?

88

Yet, why had Simms acted so strange and unfriendly that day. And for no reason Eddie could think of. There was also the memory of the strange metal tube which had been in the

rowboat when the two men went fishing, and wasn't there when they returned.

There was no point in even mentioning it to Teena, but Eddie had a strangely uncomfortable feeling on seeing the man right there on the campus from which the secret radioisotope had been stolen recently.

No, it really couldn't mean a thing, Eddie told himself.

Then, again, maybe it could.



CHAPTER SIX

Eddie had hoped that the following day he and Teena could make the trip to Cedar Point with the Geiger counter. It had been in the back of his mind ever since his dream of locating radioactive ore on the rocky point. But by the time he finished mowing his lawn and doing the few other chores lined up for him, it was too late to attempt the long trip.

Besides, the only sensible way to get to Cedar Point was by boat across Moon Bay. It was a two-mile row each way. Yet, protected by sand bars, the bay usually remained quite calm. By taking it easy, it was no great job rowing out to the point.

Still, you had to have a rowboat. To rent one cost money. With so many things going on during summer vacation, Eddie simply hadn't been able to save out of his allowance.

90

He knew of a way to get a boat, though. He had done it a couple of times before. There were more than two dozen rowboats at Anderson's Landing. Seldom were they all rented at once.

On this particular Saturday, a gray blanket of high fog hung in the sky. Eddie had an idea that quite a few of the boats would still be tied up at Anderson's Landing. Right after lunch he hurried over to Teena's house.

"How would you like to go down to Anderson's Landing," he suggested, "and see if we can't earn a day's rental on one of the boats? Then maybe next week we can take that trip out to Cedar Point with the Geiger counter."

“You want to earn the use of a rowboat again?” Teena asked.

“Yep. It’s not so hard,” Eddie said. “Want to come?”

91

“I’ll ask mother.”

Soon Eddie and Teena arrived at the beach. Mr. Anderson was midway out on the wharf which jutted a hundred or so feet out into the smooth water of the bay. They trotted out across the rough planking to see him. The boatowner was a small, wiry man with deep wrinkles around his eyes from years of squinting against the reflection of sun on water.

“Hi, there,” he greeted. “Where are your fishing poles?”

“We didn’t come to fish today, Mr. Anderson,” Eddie said. “Do you have any odd jobs we can do?”

“Need a boat?” the owner guessed.

“Yes, sir,” Eddie said. “We’d like to row out to Cedar Point one of these days.”

“Quite a row.”

“Oh, Eddie’s a good rower,” Teena said. “Sometimes I even help with one of the oars.”

“Well, now,” Mr. Anderson said, rubbing his bristly chin, “you’d need a boat almost a full day to row out to Cedar Point and back. At fifty cents an hour, that’s quite a bit of money.”

92

“We don’t have any money, Mr. Anderson,” Eddie explained. “That’s why we hoped we could work it out. Remember, I’ve done it before.”

“I remember,” Mr. Anderson said. “And I remember that you’re a pretty good worker, too.” He glanced along the wharf at the rowboats tied up to a row of cleats. “Tell you what. You

clean out what boats are in, and you've earned yourselves a day's rental on one."

Eddie counted the boats quickly. There were fourteen of them not in use. Depending upon how messy various fishermen had been, he and Teena should be able to clean them up in about three hours.

"How about it, Teena?" he asked.

"All right by me," she said.

"It's a deal, Mr. Anderson," Eddie said. "And thanks a lot."

"Just when do you figure you will want the boat?" the owner asked.

"Maybe next Saturday."

93

"All right. You do the job, and I'll save you one. Make it a good job, mind you."

They started with the boat near the far end of the wharf, and worked shoreward. They wiped off the seats with a damp rag and coiled the anchor ropes neatly near the bow. The biggest job, though, was cleaning up the junk which had gathered in the bottom of each boat during the week. There were candy wrappers, smelly chunks of old bait, snarled bits of leader, occasional fishhooks, even dried-out sandwich crusts and other odds and ends which had collected in each boat.

While they were working, two more boats returned. Eddie checked their numbers when they came in. Then, after he and Teena had finished cleaning up the fourteen, they went back and did the two new arrivals.

"Well, I'd say you've earned a boat for next Saturday," Mr. Anderson said, glancing approvingly at their work. "And

94

thanks for cleaning up those extra two that came in. They weren't actually in the bargain, you know."

"We were glad to do them," Eddie said, feeling a bit proud that they had done more than the bargain called for. "I guess we'd better be going now."

"There comes another boat," Teena said, pointing to one of the orange-and-white Anderson's Landing rowboats about a hundred yards out from the wharf.

"Well, now," Mr. Anderson said, smiling, "don't you be staying around to clean that one up. I'll take care of it."

"Teena, look," Eddie said. "It—it's those same two men we met at the cove last week. You know, the tall one called Simms and the chubby one called Roy."

"Roy Benton," Mr. Anderson said, consulting his rental slips. "He signed for the boat this morning. Second Saturday they've rented one. Hope they're steady customers. I can always use the business. Don't know how long they'll stick with it, though. They didn't catch a thing last week."

"Bet they didn't catch any this week, either," Eddie said. "Not if they fished over the sand bar again."

"I tried to tell them about good fishin' spots," Mr. Anderson said, "but they didn't seem to be listening. Didn't even ask me what kind of bait was best around here. Well, there are all kinds of fishermen. One thing I've learned in this business is not to go around giving advice when no one asks for it. Fishermen can be mighty touchy about that. Best to let them use up their own pet ideas, even if they don't catch fish."

"I can't figure why they would want to take trout rods out with them to do ocean fishing," Eddie said.

“Trout rods?” Mr. Anderson asked. “They rented poles from me. I didn’t see any trout rods.”

“Well, remember that metal tube they had last week? About two feet long? If that wasn’t a carrying case for a jointed trout rod, what else could it be?”

“I don’t recollect them having anything like that,” Mr. Anderson said thoughtfully. “And I sure would have noticed it. I helped them get loaded in the boat. All they had was a small box which I figured was their lunch. Same thing this morning. No metal tubes with knock-down trout rods or anything like that.”

96

“Let’s go, Eddie,” Teena prompted. “I’d just as soon not have to meet them again. They were pretty cranky.”

But the tall man at the oars already was maneuvering the boat clumsily up to the wharf. Mr. Anderson leaned down, took the painter from the fat man’s hand, and snubbed it to the dock cleat.

The tall man, Simms, shipped oars and turned around to hand them up to Mr. Anderson. He spotted Eddie and Teena.

“Well, so it’s you two again,” he said with no show of friendliness. “You keep turning up, and we’ll think you’re spying on us.”

“We’ve been helping Mr. Anderson,” Teena defended.

“No fish again today?” Eddie said, looking into the empty boat.

“Snagged a couple whoppers,” the portly man said, “but they got away.”

97

“Did you hook them over the sand bar?” Eddie asked.

“Why not?” Simms said sharply.

Eddie glanced at Mr. Anderson. You just couldn’t hook big ones over the sand bar. The boat owner shrugged at Eddie’s inquiring look, but he said nothing.

“We’ll get them next week, though,” the man, Roy Benton, said. “You save us a boat for next Saturday, huh?”

Mr. Anderson made a note of it.

Before Eddie nodded to Teena that they should be leaving, he noticed that there was no metal tube lying in the bottom of the boat. Had he been seeing things last Saturday? After all, even Mr. Anderson claimed the men hadn’t brought anything along except a lunch of some kind.

Eddie was quite sure it hadn’t been imagination, but he didn’t know why the vision of the round metal cylinder kept coming into his mind. And anything he couldn’t explain bothered Eddie a lot.

98

At the foot of the wharf Teena said, “It’s early yet, Eddie. Let’s take a hike up the beach, shall we? Maybe we could even go as far as the lighthouse and say hello to Cap.”

“Suits me,” Eddie agreed. He never got tired of walking along the beach. There was always something new to see and do. The fresh ocean breeze on his face and the soft sand underfoot made him feel good. Nor did he ever tire of picking pebbles off the beach and skipping them across the smooth water of the bay.

A little while later they were almost to the cove when a piece of green material caught Eddie’s gaze. It was being gently buffeted up and down on the sand by the small lapping waves. He trotted over and picked it out of the water.

“What’d you find, Eddie?” Teena called from nearby.

“Just a piece of rubber,” Eddie said, holding up the four-inch length of green material. “Looks like part of a strap off someone’s swim fins.”

99

“Boy, you’re some beachcomber,” Teena teased. “An old strap off someone’s swim fin is some treasure.”

Eddie drew back his arm and was about to throw the scrap back into the water, when some printing which was molded into the rubber caught his eye.

“Hey,” he said, looking at it closely. “It’s got some kind of foreign words on it.”

“So what?” Teena said. “I guess they make swim fins all over the world. Probably some tourist from another country brought them. There are quite a few tourists around here during the summer, you know.”

“Yeah, that’s right,” Eddie admitted, but he stuffed the scrap of rubber into his pocket and walked on.

In a little while they arrived at the cove where they had come across the two men a week earlier. Owing to the rocks and the rather poor beach, the cove was seldom visited by bathers. There was really little reason for fishermen to put into the cove, either. That was why it had puzzled him to find the two men at the cove the previous Saturday. However, they might simply have been exploring the cove.

100

Eddie and Teena continued across the rough beach. There was no one in sight at the cove. As they walked, they picked up bright shells which sprinkled the sand before them.

“Look at these tracks, Eddie,” Teena said, as she pointed down at deep grooves in the sand. They were long and wide—

the kind a boat dragged up onto the beach would leave.

“They look fresh,” Eddie said. “Couldn’t have been here more than three or four hours, or the tide would have wiped out the marks. Wonder if it was the same two fellows?”

“Funny that they would rent a boat to go fishing,” Teena said, “and then come in here to the cove first. There aren’t any sand crabs to dig for bait.”

Eddie was thinking the same thing. Then he saw footprints which led from the place where the boat had been beached to the base of the bluff rising above the cove. “Now why would they go to the foot of the bluff?” Eddie said, puzzled. “There’s nothing to see over there.”

101

Curiosity gripped him. He started following the twin sets of footprints.

“Eddie,” Teena said, “we’d better go on if we’re going to visit Captain Daniels.”

“This won’t take long,” Eddie called back over his shoulder. Teena followed as he went on toward the foot of the bluff.

“Hey, look,” Eddie said. “There’s a kind of path that zigzags up the bluff. I’ve never noticed that before.”

“It’s not much of a path,” Teena said, looking up the steeply winding trail. “And I don’t know why anyone would want to use it to get to this cove. It’s much easier coming up the beach.”

“But someone’s been using it,” Eddie said. “See how the ground’s stirred up. I can’t figure why anyone would want to land a boat in this cove, then climb up and down that bluff before going fishing. Can you?”

“Guess not,” Teena admitted.

102

“Might be worth finding out,” Eddie said. “Come on.”

“Where?”

“Up the path. I’ve never been up top. Might be something really worth seeing.”

“I doubt it,” Teena said. “There used to be some fishermen’s shacks up top. But I don’t think anyone lives there any more.”

“Just for kicks, let’s see,” Eddie insisted, starting up the winding path.

The dirt of the bluff was sandy and soft, making hard climbing as it shifted and slid underfoot. It took them several minutes to climb the slanting palisade which was only some seventy feet high.

“Phew,” Teena said, after they had scrambled up the last few feet. “You sure get some wild ideas, Eddie.”

Eddie didn’t answer. He stood looking around. It was easy to see why so few people were acquainted with that part of the coastline. Although there were no trees of any size, the rolling land which extended back from the bluff’s edge was covered with a dense tangle of brush. Only a foolish person would try clawing his way through it to get to the cove. The beach route was the easy way, as Teena had said.

103

Yet there was a faint path winding inland from the top of the bluff. It disappeared quickly into the brush. Fresh footprints indicated that it had been used recently.

“Now, why do you suppose anyone would go that way?” Eddie wondered aloud.

“I wouldn’t want to try it,” Teena said. “That brush would scratch my arms and legs.”

“Maybe it leads to that shack over there,” Eddie said, pointing.

He could see only the upper half of the small building. Probably it had once been a fisherman’s house. The other fishermen’s buildings must have rotted away and fallen into the weeds. The dampness of the seashore could rot timbers out if they weren’t kept up properly. Even the structure that still stood about two hundred yards away was badly weather-beaten and without paint. The shingles of the roof were crooked and partly blown away, leaving gaping holes.



"Come on, Eddie, let's go back."

"Well, I don't know why anyone would want to go to that place," Teena said. "Surely, no one lives there. It—it almost looks haunted. Come on, Eddie, let's go back. It's getting too late to visit Captain Daniels, anyway."

“I’d like to get a good look at that shack,” Eddie said.

“But why?” Teena insisted. “I can see enough of it from here.”

Eddie didn’t have a ready answer for wanting to look more closely at the shack. He couldn’t even explain it in his own mind, let alone give a good reason to Teena. Still, a lot of things seemed to be in need of some explanation. Why did the two men named Simms and Benton come to the cove? Why did their footprints lead up the bluff and disappear into the brush? Did they lead to the old shack? If so, why? And what about that disappearing metal cylinder which Eddie had seen in their boat last Saturday?

Maybe none of it meant a thing. Even if it did, he certainly had no idea what it was. Eddie shrugged. Sometimes his curiosity got the best of him. Anyway, why bother Teena with it?

“All right,” he said, “let’s go back.”

106

107



CHAPTER SEVEN

The following few days were sultry and hot. Eddie stayed pretty close around home. He saw little of his father. Between regular teaching duties and the search still going on for the stolen radioisotope, Mr. Taylor was very busy. Each day he looked even more tired. Eddie could only imagine how much the loss of the secret radioactive substance bothered him.

Then, Friday, something happened which set all Oceanview astir. The cause was a story on the front page of the *Globe*. There wasn't positive proof, but one of the Coast Guard planes on regular patrol the previous Saturday night had picked up a strange blip on its radar screen. By the time the plane had circled back to drop a flare and investigate, the image on the radarscope had disappeared. Upon dropping the flare, they had found nothing but the smooth water of the ocean just outside the entrance to Moon Bay.

108

The immediate belief was that the object had been a submarine. Further, if it was a submarine, it certainly had been a foreign craft. The locations of all American submarines were well charted and known by the Coast Guard.

Finally, after a week of secret investigation had revealed no proof of the object's actual identity, the story was released to the newspapers.

"Dad," Eddie said that morning after breakfast, as his father prepared to leave for school, "what would a foreign submarine be doing around here?"

“That’s a tough question to answer, Eddie,” Mr. Taylor said. “And remember, what showed up on the airplane’s radar wasn’t positively identified as a submarine. It might have been a whale. Or several whales, for that matter.”

“The newspaper doesn’t think so,” Eddie said. “Besides, no one’s ever seen whales that close in.”

“There’s always a first time.”

“But what if it was a submarine?” Eddie insisted.

“It’s possible that it got off its course and surfaced to try and get a bearing,” his father said. “If that’s the case, they probably were considerably startled to find themselves so close to shore, and dived immediately to avoid discovery. It could happen. Submarines have been known to scout off this coast. But usually they are far out to sea in international waters.”

“Maybe they were picking up spies,” Eddie blurted. “Or—or landing some.”

His father looked at him sharply. “What kind of harum-scarum talk is that, Eddie?” he demanded.

Eddie swallowed uncomfortably. He wished he hadn’t said it. But he had been doing so much thinking about the stolen radioisotope and the missing blueprints from the Acme Aviation Company that the words had leaped from his mouth without his realizing it.

Before Eddie could think of an answer, his father’s face relaxed. “Forget it, son,” he said. “You always have had a pretty active imagination. There’s nothing wrong with that. Just don’t let it get away from you. Well, I’d better be leaving.”

“Dad,” Eddie said, “do you have a teacher at school named Simms?”

“Simms?” his father replied. “I don’t recall any Simms. What department?”

“I don’t know,” Eddie said. “Teena and I saw him out fishing a couple of times with a fat man called Roy Benton. Then I thought I saw Mr. Simms last week on the college campus.”

“Well, we have nearly two thousand enrolled for summer courses, you know,” his father explained. “Many of them are adults. Teachers taking extra credit courses, or studying for their masters’ degrees. I imagine a lot of them go fishing on their days off. Any reason I should know this Simms?”

111

“I guess not,” Eddie said. He was a little embarrassed at the questions he had asked. He didn’t really know why he had asked them. Yet he felt that the various puzzling things which had happened during the past weeks might tie in together. He couldn’t explain the feeling, but it gained strength all of the time.

It was the reason, too, why he decided late that afternoon to go and take a look at the shack he and Teena had seen located back from the top edge of the bluff the previous week.

He decided not to ask Teena to go. She had worried the other day about the brush scratching her arms and legs. It would be simpler to go by himself. He decided to take Sandy along for company.

He stopped at Anderson’s Landing long enough to check with the owner about a boat for him and Teena the next day.

“That’s our agreement,” Mr. Anderson said, smiling. “After all, you earned it. Don’t want you chasing any submarines with it, though.” The boatowner laughed. Apparently people

112

weren't taking the rumored submarine sighting very seriously. Eddie supposed that, as long as there was no proof, perhaps it was just as well. Besides, even a foreign submarine was not likely to cause any trouble. After all, there was no war going on.

Still, Eddie couldn't shrug it off so lightly. The tangle of strange happenings during the past days upset him, and he didn't feel much like joking; not when his father and Teena's father were both in the thick of serious trouble.

Eddie took his time getting to the cove. Sandy chased back and forth into the surf after bits of driftwood which he kept dropping at Eddie's feet, and which Eddie threw back into the water.

By the time he reached the cove, Eddie wished he hadn't dawdled along so slowly. The sun had dropped fast, and was already squashing down against the horizon.

"Come on, Sandy," he said, starting for the foot of the bluff. "We've got to hurry."

113

He started up the narrow winding trail. Sandy scurried ahead and finally stood, panting heavily, on top of the bluff, waiting for Eddie.

The shack was still plainly visible in the waning light. Eddie started along the path. In most places it could hardly be called a path, except that there were dim tracks to follow. The heavy growth of brush and weeds tore at his clothes. He kept his arms tucked in close to his body to keep from getting scratched. Sandy had no difficulty whatsoever in racing back and forth through the thick scrubby growth. All of his running had tired the cocker spaniel enough that he wasn't yipping and barking as he so often did.

Within a few minutes Eddie was to be very thankful for that.

As he had suspected, the faint trail ended at the door of the old abandoned fisherman's shack. In the eerie light of dusk, Eddie remembered Teena saying that it looked almost haunted. It certainly did. Broken shutters dangled from boarded-up windows. Gaping holes in the roof yawned at the darkening sky. The warped and twisted wooden siding made the whole structure look as though it were about to cave in.

Eddie approached the shack cautiously. He figured his curiosity would be satisfied if he took just one look inside.

His hand was poised over the latch on the door when a slight scratching sound from inside froze it in mid-air. It sounded like someone scratching a match.

Even as he stood there with sudden fear prickling along his spine, a small flare of light seeped through one of the cracks between the warped boards of the door. *It was a match!* Eddie sucked in his breath and drew back. His first thought was to turn and run.

On second thought, however, he paused. Perhaps hoboes now and then used the abandoned shack for sleeping quarters. It couldn't be very comfortable, but it would be better than sleeping outside in the damp ocean air. Although Eddie had no desire to meet any hobo, it was hardly reason to run away in a panic.

Without making any sound, and glad that Sandy was off exploring in the brush, Eddie sought one of the larger cracks in the door. Leaning toward it, he put one eye to the crack.

It was then that Eddie's fear took a firm grip on him. A small candle burned on an empty fruit crate standing in the middle of the shack's single room. In one corner was an old double bunk, empty now of mattresses or bedding. A couple of rickety chairs and a bench completed what furniture was inside the shack.

Eddie's eye was attracted by the glint of candlelight upon metal. Squinting through the crack, he was able to make out the form of the reflecting object. It was one of those metal tubes—like the one he had noticed in the bottom of the strangers' rowboat that day at the cove. On the floor was a square battery camp lantern such as hunters often use.

There was one person in the room. He sat on the small bench. His back was partly turned toward Eddie. He appeared to be studying some kind of a paper, although Eddie could see only a small corner of it.

116

There was no mistaking the man, although his face was turned away. It was the chubby fellow named Roy Benton.

There was nothing more to see. Eddie backed carefully away from the door. A few yards away, he turned and scrambled back along the darkening path toward the cove, as Sandy came crashing through the brush to meet him.

117



CHAPTER EIGHT

A lot of trouble might have been saved if things had worked out as Eddie had planned. And, yet, if they had, the mystery of the missing blueprints and the stolen radioisotope might never have been cleared up.

Hurrying home through the darkness, Eddie went over in his mind the story he would tell his father. Perhaps it didn't mean a thing. Perhaps his imagination simply was running wild, as his father had hinted. He had to admit to himself that he was prone to build rather normal incidents into deep mysteries. He had always been that way.

Even allowing for that, however, Eddie still believed there was a strong possibility that the events of the past weeks might tie in with the stolen radioisotope; perhaps even with the missing blueprints from Acme Aviation.

118

Right in the middle of those events the figures of the two men—Simms and Roy Benton—kept looming up in his mind. Anyway, he thought he should tell his father about it and let him decide whether there could be any possible connection.

It was not, however, to be that way. Upon arriving home well after dark, Eddie found his mother both irked and worried over his late return.

“You didn't even ask me if you could go,” she scolded. “And you know better than to be getting home at this late hour.”

“I—I’m sorry, Mom,” Eddie said meekly. “I didn’t know I would be gone so long.”

Mrs. Taylor turned from the stove where she was warming his dinner. “After you eat,” she said firmly, “I want you to go right to bed. No television.”

It was a mild enough punishment, Eddie thought, and didn’t argue. His father would not have been so lenient. He looked around. “Where’s Dad?” he asked.

119

“Your father phoned a while ago,” his mother explained. “He’ll be home late. Feed Sandy now; then wash up for dinner.”

Eddie opened a can of dog food, went outside, and spooned half of it into Sandy’s dish. Capping the can with a plastic cover, he put it in the refrigerator, then went to wash.

After dinner he kissed his mother good night and went straight to his room. He lay in bed, going over in his mind the recent events. He listened for his father’s arrival. He had hoped somehow to evade his early-bedtime punishment long enough to tell his story to his father. But he hadn’t counted on his tiredness. He fell fast asleep long before his father came home.

The sound of the car backing out of the driveway awakened Eddie the following morning. He washed and dressed quickly. Perhaps his mother had gone to do a bit of early shopping. It was Saturday. Probably his father would be home for the day. Now might be Eddie’s best chance to tell him what had been running through his mind.

120

Hurrying into the kitchen, he found his mother at the dinette table having a cup of coffee.

“Hi, Mom. Where’s Dad?” Eddie asked.

“He just drove out, Eddie,” his mother said. “He had to get over to school early.”

“But it’s Saturday.”

“Saturdays haven’t been very restful for your father lately, have they?” his mother said. “He has an appointment with some people from Washington D.C. this morning.”

“About the stolen radioisotope?” Eddie wondered aloud.

“Might be,” his mother said. “He didn’t say, and I didn’t ask.”

“Didn’t ask?”

“Eddie,” his mother said firmly, “the theft of that isotope is pretty serious business. Your father is handling it the best he can. He’ll tell us what he wants us to know. It is not our part to be asking questions. You try to remember that, dear.”

121

Eddie didn’t say anything. He knew his mother was right. He was greatly disappointed, though, that he hadn’t had the chance to talk to his father.

Eddie was finishing breakfast when Teena telephoned.

“We’re going to Cedar Point today, aren’t we?” she asked.

“I—I guess so,” Eddie said.

“You guess so? But Mr. Anderson is holding a boat for us. This is Saturday, you know.”

“I know.”

“Well, we should get started,” Teena insisted. “It’s some row out to Cedar Point.”

“Hold on a minute,” Eddie said. “I’ve got to ask Mom.” He turned and explained their plans to his mother.

“It’s all right, Eddie,” Mrs. Taylor said. “But no getting home late like yesterday, understand?”

“Don’t worry, Mom,” Eddie promised. “We’ll start early, and get back early.” Then he spoke to Teena.

122

“I’ll be right over,” Teena replied, “soon as I make us a little picnic lunch.”

Eddie got the Geiger counter out of the closet. He told his mother how he had been wanting to try it out on Cedar Point. He told her about the dream he had had.

“Well, I guess you never can tell,” Mrs. Taylor said. “According to your father, uranium ore often shows up in most unlikely places.”

“Cedar Point isn’t so unlikely, Mom,” Eddie said. “There are lots of rock formations out there. Uranium ore is usually located where there are plenty of rocks.”

When Teena arrived Eddie was all set to leave. He wore swimming trunks under his blue jeans. Teena said she also had her bathing suit on in case they had to swim home.

“Well, you children be careful,” Eddie’s mother cautioned. “The bay’s usually nice and smooth, but it can get pretty choppy. If it does, you head straight for shore, understand?”

123

“We will, Mrs. Taylor,” Teena promised.

Eddie’s mother smiled. “Be back no later than four o’clock,” she said to Eddie.

“Right.”

“But you don’t have a watch, Eddie,” Teena said.

“I can tell by the sun,” he said proudly, then added, “pretty close, anyway.”

Eddie carried the Geiger counter and Teena the picnic lunch as they started toward Anderson’s Landing.

Although it was a nice sunny day for fishing, there were still several rowboats tied up at the landing. Mr. Anderson was in his little office at the foot of the dock.

“I’ve been expecting you,” he said. “Saved you a nice light pair of oars, too.”

“Swell,” Eddie said. “Thanks, Mr. Anderson.”

“Take boat Number Eighteen,” the owner said. “She rides high and is leakproof. What’s that gadget you’ve got there?”

“It’s a Geiger counter, Mr. Anderson,” Eddie said. “We hope we’ll find some signs of uranium out on Cedar Point.”

124

“That the stuff you make atom bombs of?” Mr. Anderson said, with a note of disapproval in his voice.

“You can make bombs of it,” Eddie admitted, “but nowadays scientists are more interested in running machinery and curing diseases with it.”

“That’s the kind of thing I like to hear,” Mr. Anderson said, smiling. “In that case, good luck.”

Soon, with Eddie at the oars, they started toward distant Cedar Point. Eddie set the course in a line which cut at an angle across the bay. As they were crossing the submerged sand bar, Teena pointed shoreward.

“Eddie, look,” she said.

Resting on the oars and following the direction of Teena's finger, Eddie saw that they were directly offshore from the cove. He also saw the orange-and-white rowboat pulled up onto the beach.

Of even more importance, he saw two men making their way carefully down the narrow trail which zigzagged down the face of the bluff. Even from the distance, Eddie saw that one man was tall and thin, the other short and fat.

125

"It's those two men!" he exclaimed.

"Of course. But don't get so excited," Teena said. "They're probably just coming out fishing."

"Coming from where?" Eddie asked. He knew there was only one place to come from—the shack. Then he remembered that Teena didn't know about his visit to the shack yesterday.

"Well, anyway," Teena said, "let's get away from here before they come out to fish over this sand bar. They sure would think we were spying on them if they found us out here. I hope they don't see us now."

"To far away for them to tell who we are," Eddie assured, sharing Teena's dislike of meeting the two men again.

Eddie started to turn back to rowing, when the flash of sun on metal caught his eye. He knew at once that one of the men was carrying that metal cylinder which he had seen yesterday evening in the shack, and which he had puzzled over so long. He would like to have stayed and gotten another look at it; that is, if the two men were coming out to fish over the sand bar again. Yet Teena's warning about getting away seemed the wiser move. Eddie bent to the oars.

126

Less than an hour later he guided the boat onto the narrow beach at Cedar Point.

“Phew!” he said, mopping the sweat from his forehead. “That’s a lot of rowing.”

“It was a swell ride, Eddie,” Teena said. “I’ll row back if you want.”

“You’re a girl,” Eddie said importantly, which seemed to close the subject about Teena doing the rowing. But Teena did help him drag the boat up onto the beach beyond the high-water mark.

“Now to find some uranium,” Eddie said, picking up the Geiger counter. Before starting inland to explore the point, however, he shaded his eyes and looked back across the bay. In the far distance he could barely make out Anderson’s Landing. Quite a few boats dotted the bay in between. Directly in line between Cedar Point and Anderson’s Landing was the light strip of water marking the submerged sand bar. There was only one boat over the sand bar.

127

“Those two fellows are fishing in that same place again today,” Eddie said. “They don’t seem to learn, do they?”

“Let’s not worry about them,” Teena said. “Let’s start prospecting. We promised to be home by four. It’s a long trip back.”

The wind-swept point offered difficult hiking. Fallen trees and tangles of underbrush slowed their progress. They had to keep on the lookout for poison ivy.

“If leaves there are three, leave it be,” Eddie said, remembering the familiar warning. They gave wide berth to the irritating vine whenever they saw it.

Eddie left the Geiger counter switched on much of the time. The way led over the rocks. There was no way of telling, except by the Geiger counter, if any of the rocks were

128

radioactive. The results, however, were quite discouraging. Except for the faint background count, the Geiger counter gave no sign of there being any uranium-bearing ore on Cedar Point.

After a tiring hour and a half of hard climbing over and around the outcroppings, Teena suggested they stop and eat their picnic lunch.

“Might as well,” Eddie said. “Sure doesn’t look like we’re going to find anything out here. Lot of trouble for nothing, huh?”

“Oh, no, Eddie,” Teena disagreed. “We’re having fun aren’t we? After all, you’re supposed to have fun during vacation.”

“Be better, though, to have fun and find some uranium, too,” Eddie said.

Teena laughed. “You sure do want everything,” she remarked.

Eddie switched off the Geiger counter. They found some shade under a wind-twisted oak and ate their lunch. Eddie glanced at the sun. “It must be one o’clock,” he said. “Guess we’d better be starting back. The water will be a little choppy than this morning. Won’t be so easy to row. I don’t want to get home late, or my mom will scalp me.”

129

“Let’s go,” Teena said. “Anyway, we’ve done enough prospecting out here to know there’s no uranium around.”

On the way back to the boat Eddie tried out the Geiger counter in a couple of places they had missed. The results were the same—negative. He put the Geiger counter into the bottom of the boat, pushed the boat into the water, and jumped in after Teena.

A slight breeze angling in over the bow made rowing difficult. Less than halfway across the bay, Eddie's arms and shoulders began to ache.

"Eddie," Teena said from her seat in the stern, "why don't I sit there beside you and row with one of the oars? We've done it before. Just give me a little time to get the swing of it."

"O.K.," Eddie said tiredly.

Teena moved up beside him on the wide center seat and took the starboard oar handle in both hands. After a couple minutes of splashing and going in circles, they settled down to pulling together smoothly and evenly.

130

"Hey, this is the best deal," Eddie admitted.

"You see," Teena said, "even girls can be some help."

Eddie smiled. As they were approaching the near end of the under water sand bar, he craned his neck around.

"Good," he said, "those two men have gone, so we won't run across them again."

They kept pulling together. The water turned light in color as the sun reflected off the yellowish sand lying a scant ten feet beneath the surface.

Suddenly Teena stopped rowing. "Eddie," she said, "what's that?"

"What's what?" Eddie asked, resting on his oar.

"That clicking."

"Clicking?"

He heard it then, almost beneath the seat. He glanced down.

“Yipes!” he exclaimed. “I—I forgot to switch off the Geiger counter before I put it in the boat. Boy, if the battery is weak, Dad’ll—”

131

He reached toward the switch. His hand stopped in mid-air. The needle of the radioactivity gauge was quivering far over to the right, and the clicking which had attracted Teena’s attention was much louder and faster than the normal background count.

“Teena! There—there’s radioactivity around here!”

“In the water?”

“No. Maybe underneath the water. Maybe on the sand bar. This is a sensitive Geiger counter. It could pick it up all right.”

“Hey, the clicking’s getting weak again,” Teena said.

“We must be drifting away from whatever is causing it,” Eddie said. He moved the Geiger counter up onto the seat between them. He put the earphones on. “Now let’s kind of circle around here and try to pick it up again.”

With nothing but a broad expanse of water and no marker to guide them, trying to locate the spot where the Geiger counter had sputtered to life was anything but easy.

132

“Eddie, I think we’re getting farther away all the time,” Teena said ten minutes later.

“But if we don’t find it now we might never find it again,” Eddie said. “Just a little more. Pull easy on your oar. We’ll circle to the left and—Hey, there it is!”

The rapid clicking through the headset filled his ears. “Hold ’er steady,” he said. He crawled quickly to the bow of the

boat, lifted the heavy concrete anchor over the gunwale, and eased it down onto the sand bar with the Manila line attached.

“There. We won’t lose it now,” he said.

“Lose what, Eddie?”

“Whatever’s making the Geiger counter act up,” Eddie said.

“This would be an awful wet place to have to mine for uranium,” Teena said.

“It could be done,” Eddie insisted. “Boy, we must be right over it. Listen to those clicks. And look at that needle jump around.”

Teena looked over the side. “It looks to me like plain old yellow sand down there,” she said.

133

“Might be some uranium-bearing rock under it,” Eddie said. He leaned over his side of the boat. Although the sand bar was not far below, the water was somewhat murky, and the ripples on the surface made it difficult to see anything on the bottom. “Might be a tough job getting at it, all right, but—”

The rest of the words died in Eddie’s throat, as a glint of metal flashed in his eyes.

“Teena, there—there’s something down on the sand bar!”

“What do you mean, something?”

“Something bright. Like metal.” Eddie put his face as close to the water as he could without falling out of the boat. “I can see it now!” he exclaimed. “It’s about two feet long. Two or three inches thick. It looks round, and—”

“Eddie!” Teena said. “The metal tube you saw that day in the rowboat. You know, the day we came across those two men at

the cove. Remember?”

“I remember,” Eddie said, for the thought already had sprung into his mind.

134

Now two other thoughts crowded in behind it. Both were puzzling thoughts which left his mind reeling.

What was the metal tube doing there below on the sand bar? Why, above all things, was it sending out radioactive rays?

135



CHAPTER NINE

“Eddie,” Teena said, “what are you going to do?”

Already Eddie had pulled off his shoes and T shirt. He slipped off his blue jeans, and stood in his bathing trunks ready to dive overboard.

“I’m going down and get that thing,” he said.

“Why?”

“Something’s mighty crazy about all this,” Eddie said.

“Maybe so,” Teena agreed. “But Eddie, isn’t it dangerous? If that thing’s radioactive—”

136

“No. It’s not that radioactive,” Eddie said. “Those two men handled it all right. It’s some special kind of a tube. I’m going to take it to my dad.”

“Why?”

“Why!” Eddie repeated impatiently. “Because it must have something to do with that stolen radioisotope, that’s why.”

“Well, you don’t need to get sore at me,” Teena scolded.

“I’m not, Teena,” Eddie said, calming down. “But I’m not going to take the chance of leaving the thing out here and maybe not being able to find it again.”

“You sure it won’t hurt you?” Teena asked again.

“I’m real sure,” Eddie said. “I’ve seen that Geiger counter act up a lot worse over a small sample of uranium ore. Don’t worry. If it was really hot with radiation, those two men wouldn’t have been handling it either. Dad said that whoever stole the isotope would be an expert on knowing how to handle it safely. I’ll be right back.”



The cylinder was simple to locate.

Eddie slipped over the side and into the water. He took a breath, nosed over, and kicked downward. It was a shallow dive compared to some he had made while skin diving for lobster. He stroked easily down to the sand bar. The bright metal cylinder was simple to locate even in the murky water. He grabbed it with his left hand and swam back to the surface.

Teena took it from him and laid it in the bottom of the boat. Then she helped him climb in over the stern.

“It doesn’t look like anything very special, does it?” she said.

“That’s why I thought it was for holding a jointed trout rod,” Eddie said, “but watch this.” He switched on the Geiger counter and held the probe near the metal tube. The earphones began to sputter with continuous clicks. The indicator on the dial jumped far forward.

“It’s a cinch there’s some radioactive stuff inside,” Eddie assured. “Let’s go. I’ve got to take this to my father.”

As they drew within a quarter of a mile of Anderson’s Landing, Teena said, “Eddie, what if those two men should be around the boat dock and see us with this tube?”

Eddie looked over his shoulder. He saw only one person on the landing. That would probably be Mr. Anderson.

“I’m sure they’ve already turned in their boat and gone on home,” Eddie said, “but just to be safe, we’ll do this.” He picked up his blue jeans which he had left wadded up in the bottom of the boat, as he hadn’t wanted to put them back on over his wet trunks. He pushed the metal tube into one of the empty legs. Then he wrapped the excess material around it. “There,” he said, satisfied, “no one can see it now.”

Mr. Anderson came out to meet them as they eased the rowboat gently up to the dock.

“Well, how did it go?” he asked. “Find any atoms?”

Eddie smiled. “Everything is made of atoms, Mr. Anderson,” he explained. “We were looking for uranium. That’s a special kind.”

140

“I guess it’s a special, all right,” the boat-owner agreed, “the way it can blow things to smithereens.”

Eddie didn’t argue, but he wished people would stop thinking that all radioactive materials were used to blow things up. He supposed, however, that since the atomic bombs were what really started what came to be called the Atomic Age, it would take some time to educate the public to the fact that atomic power was a much greater builder than a destroyer. Anyway, at the moment he didn’t want to get into a long discussion about it.

“We didn’t find any uranium, Mr. Anderson,” he said.

“But we had a swell trip,” Teena put in. “Thanks for letting us use the boat.”

“You earned it,” Mr. Anderson reminded them. “By the way, those two fellows who have been fishing over the sand bar came in a while ago. Didn’t catch a thing. Sure a stubborn pair, aren’t they?”

“Maybe they just don’t like to clean fish,” Eddie said. But he was glad Mr. Anderson had mentioned the men. It added evidence to his belief that they weren’t the least bit interested in fishing, anyway.

141

After cleaning up the boat, he and Teena started along the dock. Eddie carried the metal tube rolled up in his blue jeans.

“Any time you want a boat,” Mr. Anderson said, as they stepped off the dock and started across the beach toward

home, “you’re always welcome to earn it the same way.”

“Thanks,” Eddie called back. “We may need one again before long.”

It was a little past three o’clock when they reached Eddie’s house.

“You certainly made it in good time,” his mother said. “Any luck?”

“Not at Cedar Point,” Eddie said. “But, Mom, we found something else. Where’s Dad?”

“He’s not home from school yet,” his mother said. “What’s that you have wrapped up in your jeans?”

Eddie told her quickly, without going into all of the background.

142

“You think it has something to do with the stolen radioisotope?” his mother asked in disbelief, when he had finished.

“I don’t know, Mom,” Eddie said. “But why would it be radioactive?”

“You haven’t opened it, have you?”

“No. It’s sealed tight,” Eddie said. “I—I thought Dad should do that.”

“You’re right. You run it over to school and find your father.”

Teena spoke up for the first time. “Eddie, if that tube really belongs out there and we took it, we—we might get in trouble.”

“Belongs out there?” Eddie asked.

“Maybe the Coast Guard is using it for some kind of a test or something,” Teena said.

That was a possibility which hadn’t occurred to Eddie, yet he quickly dismissed it from his mind. The two men who had planted it out on the sand bar certainly had nothing to do with the Coast Guard or anything like that. Nor would it have been in the shack yesterday evening.

“Not a chance,” he said. “Anyway, I’m going to take it over for Dad to see.”

143

“I’ll call him and tell him you’re on your way,” his mother said.

“You want to go along, Teena?” Eddie asked.

“What a question,” Teena said. “Sure, I want to go.”

“Eddie,” his mother reminded him, “you can’t go over to school in your swim trunks. Go slip on some denims.”

Eddie hurried to his room and put on some freshly laundered denims. Then, leaving the metal tube still wrapped in the blue jeans, he and Teena started down the street toward the college campus.

Mr. Taylor was waiting for them in front of the nuclear-science building. He seemed strangely excited. Eddie wondered what his mother had said over the telephone.

“Let me take it, son,” Mr. Taylor said, reaching out for the blue jeans in which the metal cylinder was wrapped. He turned to go inside.

144

“Can we come with you, Dad?” Eddie asked quickly.

“Of course, of course,” his father said over his shoulder. “Come along. If this is anything like your mother said, there’ll

be a lot of questions to ask.”

Eddie’s father led them through his office and on into a dressing room where they pulled on specially treated white coveralls, gloves, and hoods which fitted over their heads. Each hood had a small glass window for looking out.

“Just an extra precaution,” Eddie’s father said. “Really not necessary, but we simply don’t take any chances with possible stray radiation.”

They went on into the large laboratory. Eddie had been there before. The sight of the fantastically shaped apparatus used in various atomic-research tests always excited him.

There were several men in the room. Each was dressed in white coverall-type protective suits similar to those he and Teena and Mr. Taylor wore.

145

In the center of the laboratory stood a square booth with thick walls and a glass window in the front wall. Eddie knew the walls were lead-lined, and the glass was a thick, specially treated type. All experiments which were the least bit hazardous were conducted inside of that six-by-six-foot booth. The radioactive materials were handled remotely by a strange steel-fingered device operated by a man who stood safely outside of the booth. Absolutely no chances were taken in the handling of radioactive materials.

Eddie’s father inspected the tube closely, as he went toward one of the many complex devices that filled the laboratory.

“It’s a careful job of machining on this tube,” he said. “Surely not the work of amateurs. Seems to be a lead alloy of some kind. Probably worked out in thickness and amount of lead in the alloy so as to allow just the right amount of radioactive rays to leak through without being dangerous.”

He flicked several switches and turned various knobs on the instrument under which he had placed the tube. Eddie watched dial needles quiver and lights flash, wishing he knew what they meant.

“All right,” his father said, turning off the machine, “you’re exactly right. There’s radioactivity inside that tube. Plenty of it, I imagine. Yet, only enough of it is allowed to leak out to furnish a tracer. It was a regular beacon leading you right to it with your Geiger counter.”

“Dad, you mean—”

“Let’s hold the questions a while, Eddie,” Mr. Taylor interrupted. “We’ve got a few tests to run on this first. There are some things we need to find out for sure.” He called to one of the young men working at the far side of the room. They talked for a few moments while the laboratory worker inspected the cylinder closely.

Then he took it inside the shielded booth and laid it on the table beneath the strange contraption with the protruding metal arms and pincers. Several other pieces of testing apparatus were placed on the table. Then he came back outside, closing the door carefully behind him.

“All right, Mr. Taylor,” the young technician said, “we’ll see what we can do with it.” He slipped his hands into the grips which operated the metal fingers on the far side of the thick, protective glass through which they watched.

Eddie and Teena looked on fascinated as, controlled from outside, the mechanical clamps on the metal arms inside picked up the tube. Then wrenchlike metal fingers wrapped around one sealed end. After much twisting and prying, the tight fitted cap came off.

“So far, so good,” the young scientist said. “Now let’s see what’s inside.” He moved his own hands and the mechanical fingers inside tipped the tube on its end. A small black capsule slid out onto the table. It was about the size of a dime-store beanshooter.

The metal fingers kept working until the cap sealing the small black capsule was removed. When it was tipped on end a yellowish powder trickled out into a small bowl which had been placed on the table inside.

148

The metal fingers continued working. They placed the small bowl with the yellow contents under one instrument after another. Knobs were turned and readings were jotted down. After the final test was made, Eddie’s father studied the results carefully. He compared them with the formulas on a piece of paper he had brought from his office.

While waiting silently, Eddie’s gaze went back to the large uncapped silver-gray cylinder still lying inside on the table. What appeared to be a corner of a sheet of paper jutted slightly out of the open end.

“Looks like there’s something else inside of that tube,” he said to the young technician beside him. Talking beneath his hood muffled his words, yet the scientist seemed to have no trouble understanding.

“By George, you’re right,” he said. He reached once more for the proper grip rings and levers to operate the robot fingers inside. “Let’s see what it is.”

149

He tipped the tube so the open end was down, then shook it. A large piece of rolled-up paper dropped out. As it fell to the table, it unrolled part way—enough, at least, for Eddie to see the blue color of its inside surface. He also saw the white markings.

“Blueprints!” he cried.

At the word, his father looked up from his own busy figuring. “You’re right,” he said. “They sure are blueprints. You kids certainly hit upon something big. Mighty big.”

“What do you mean, Dad?” Eddie wondered.

“There’s no doubt about it,” his father said firmly. “The material that came out of the small black capsule inside of that tube is a part of the stolen radioisotope. It’s mixed in with some other material to weaken its power. But I’m certain the radioactivity comes from small amounts of our isotope.”

“Then we’ve found the stolen isotope!” Eddie exclaimed. Although the idea had occurred to him before, hearing the proof of it was no less startling.

150

“Only part of it,” his father reminded him. Then he turned toward Teena. “Unless I miss my guess, those blueprints are some of the ones missing from Acme Aircraft Company.”

This seemed sheer fantasy, like something that might happen in a restless dream after eating too much ice cream and lobster salad.

“Come on, kids,” Mr. Taylor prompted, leading them back toward his office. “There’s a lot to be done. And unless I miss my guess, it must be done quickly—or we might be too late.”

151



CHAPTER TEN

Once back inside of his office, Mr. Taylor motioned for Eddie and Teena to be seated. Then he picked up the phone and made two quick calls. They also must have been local calls, Eddie thought, for within five minutes two men hurried into the office. Both were dressed in normal summer business clothes.

Eddie's father introduced the dark-haired one in the light-tan suit as Mr. Paul Evans of the Federal Bureau of Investigation. The other man was tall, light-haired, and blue-eyed. His name was Walter Jamison. He was from the Drake Ridge atomic reactor. Eddie's father didn't explain what either man was doing there, but Eddie had no doubts that their main interest was recovery of the stolen radioisotope. Probably they had been around the Oceanview College campus ever since the theft had taken place.

152

"All right, Eddie," Mr. Taylor said after the two men sat down. "Start from the beginning and give us the whole story. Don't leave anything out. Teena, you see that he doesn't."

Eddie didn't know exactly what his father meant by the whole story. But he started with the day when he and Teena had come upon the two men at the cove. He told about their somewhat strange actions, and the puzzling sight of the sealed metal tube lying in the bottom of the rowboat. He mentioned how the men had not had it with them when they returned the boat to Anderson's Landing a while later.

He told about seeing the men fishing out over the sand bar the Saturday after that, and again today.

Then, to Teena's surprise, he brought in yesterday's lone hike out to the cove. He told of his curiosity over the tracks leading to the abandoned fisherman's shack set back from the top of the bluff, and how he had been greatly surprised, at peeking through the crack in the door, to see the chubby man named Roy Benton inside, as well as a bright metal cylinder—like the one they had just taken apart in the laboratory—standing in a corner of the shack.

153

“Probably it was the same tube you just took apart,” Eddie said. “When Teena and I were rowing out to Cedar Point this morning, we saw the two men coming down the bluff carrying something shiny. Later, looking back from Cedar Point, we saw them anchored over the sand bar. Probably over the same place we found the cylinder.”

“It figures,” Mr. Evans, the FBI man said. “For the sake of argument, let's say the two men are spies. Could be even more than two here at the college or working at Acme Aircraft.”

“Spies!” Teena gasped in disbelief.

“Maybe they're both hiding at the shack,” Eddie said excitedly. “You'd better go arrest them!”

154

“Not so fast,” Mr. Evans said. “Arresting them isn't nearly so important as finding out where the remainder of the radioisotope is hidden. Getting hold of the rest of those missing blueprints also is much more important than arresting two men.”

“In fact,” Mr. Jamison added, “arresting them too early might tip off the whole operation, and everyone would run for cover before we could pin anything down.”

Just then Teena's father came hurrying into the office. "Sorry I couldn't get here sooner, Steve," he said to Mr. Taylor, "but we were trying to locate another very important set of blueprints. More secret guidance-system parts. I absolutely can't figure how those blueprints can keep disappearing, and —"

Eddie's father held out the rolled-up blueprints which had been inside the metal tube. "These wouldn't happen to be the ones, would they, Tom?" he said.

One glance, and Mr. Ross's face took on an expression of mixed pleasure and amazement. "They certainly are!" he exclaimed. "But how—"

155

The FBI man interrupted. He brought Teena's father up to date on the story thus far. Mr. Ross looked over toward Teena and Eddie. As pleased as he seemed over the recovery of at least part of the missing blueprints, he appeared even more concerned over something else.

"If I had had any idea that you two were getting mixed up in anything like this," he said, "I'd have insisted that you stay home and play scrabble or checkers or something safe."

"We—we weren't mixed up in anything, Mr. Ross," Eddie said quickly. "At least, we sure didn't know we were, and—"

"I believe," the FBI man cut in, "that we'd better get down to cases. We may not have much time to solve this problem. Let's see what we have to go on thus far. Then we'll try to plan our next move."

Eddie listened as Mr. Evans reviewed the situation point by point. Two men—the one called Simms and the other known as Roy Benton—were involved in stealing the blueprints and the radioisotope. Mr. Evans didn't seem at all worried about capturing them when the time was ripe. On each of the last

156

three Saturdays, including today, Eddie and Teena had seen them fishing, or pretending to be fishing, over the sand bar in Moon Bay.

“We might assume, then,” the FBI man said, “that on the past two Saturdays the men’s real purpose for going out in the boat was to drop other metal tubes overboard. Other tubes similar to this one.”

“And remember,” Mr. Taylor said, “the first time Eddie and Teena saw them was the very Saturday after the isotope was stolen from the college.”

“Right,” Mr. Jamison said.

“So,” the FBI man picked up the line of thought, “the question is why the men dropped the metal tubes out on the sand bar. It’s a fairly safe bet that each tube contained a little of the radioactive material, plus other blueprints. Let’s assume that the reason behind the whole thing is to smuggle the blueprints out of the country.”

157

“That would go for the isotope, too,” the man from Drake Ridge said. “It was a new secret isotope, you know. Various foreign governments would like to get their hands on it.”

“But the men didn’t talk like foreigners,” Eddie said.

“Of course, they wouldn’t,” Mr. Evans said. “Might not even be foreigners. Unfortunately, there are a few greedy people who will do almost anything for money.”

“Even spy?” Teena said, aghast.

“Even spy,” Mr. Evans said. “But what we need to find out is how anyone is managing to smuggle the stuff out of here.”

“Probably by boat,” Mr. Jamison said.

“The Coast Guard keeps close tabs on all boating,” Mr. Evans said. “And the bay’s too shallow to allow ocean-going ships inside.”

A thought sprang into Eddie’s mind. “Mr. Evans,” he said, “I found a rubber strap on the beach last week. It looked like a strap broken off a swim fin or something like that. It—it has some foreign printing on it. I have it at home.”

158

His announcement had an immediate effect. “That should give us a real clue,” the FBI man said quickly. “It makes sense, too, that the cylinders would be recovered by skin divers. Perhaps foreign divers similar to our own frogmen.”

“It would have to be done after dark,” Mr. Ross said. “Otherwise they would be seen. And how could anyone locate a small cylinder like that under ten feet of water at night.”

“I think I can answer that,” Eddie’s father said. “In fact, Eddie and Teena found that answer. It could be located with a Geiger counter.”

“That’s it,” the FBI man agreed. “For instance, they could use a rubber boat to sneak in under cover of darkness. They would know the approximate location of the cylinder.”

“How?” Teena’s father asked.

“By some established plan. Probably by triangulation. They could use the lighthouse for one reference point. Perhaps some other signal light on shore would give them a second point.”

159

Quickly, Eddie told him about seeing the heavy-duty battery lantern in the shack. “They might use it for a signal light,” he said.

“Very possible,” Mr. Evans agreed. “Anyway, a little quick figuring would locate the spot on the sand bar where the men had dropped the cylinder. A Geiger counter could pinpoint it quickly. The diver would recover the cylinder, climb into his rubber boat, and paddle back out—” His words dwindled away to thoughtfulness.

“Paddle back out to what?” The man from Drake Ridge voiced the thought that was in all their minds.

Eddie wondered if the same answer that immediately occurred to him was shared by the others. Although soon after the article had appeared in the Oceanview newspapers most readers had discarded it as nothing more than an unfounded rumor, Eddie had never quite forgotten it. Nor had the Coast Guard officially withdrawn its belief of what had been sighted by its radar equipment that Saturday night two weeks ago.

Now there seemed no argument. It was, in fact, the only logical method by which the isotope and the plans could be smuggled away without detection.

“The submarine!” Eddie exclaimed.

160

161



CHAPTER ELEVEN

The parts of the mysterious jigsaw puzzle had begun to fit into a rough pattern, with Eddie and Teena furnishing most of the key pieces. Mr. Evans glanced at his wrist watch.

“It’s now a quarter after five,” he said. “The supposed submarine sighting took place on a Saturday night two weeks ago. That same day Eddie and Teena saw the men out over the sand bar. They saw them again last Saturday. Probably another pickup took place that night. It’s logical, therefore, to assume that the third pickup is scheduled for tonight. That doesn’t give us much time to set our trap.”

“You’re the boss,” Mr. Taylor said. “You tell us what you want us to do.”

162

“That’s right,” Teena’s father said anxiously. “I can’t overemphasize how important it is that those blueprints don’t get out of this country.”

“First I have several urgent phone calls to make,” the FBI man said quickly. “Must get the wheels turning at once.”

“There’s a phone in the empty office next door,” Mr. Taylor volunteered. “Help yourself.”

While the federal investigator was in the next room telephoning, Mr. Taylor, Mr. Ross, and the man from Drake Ridge talked over what they knew so far.

“Apparently what we have to go on,” Eddie’s father said, “are some assorted guesses, none of which may prove to be positive facts.”

“Well, guesses will have to do for the moment,” Mr. Ross said. “We have to have a starting point.”

“All right,” Mr. Taylor agreed, “here’s what we have. Two men seem responsible for both the stolen isotope and the missing blueprints. Eddie and Teena both saw the tall one called Simms on the college campus about a week ago. He must be familiar with our atomic-research department in order to know of the delivery, and to plan a method for stealing the isotope. In that case, he shouldn’t be difficult to trace.”

163

“Dad,” Eddie said suddenly, “doesn’t everyone who works around the atomic lab have an identification badge with his picture on it?”

“You’re absolutely right,” his father said, getting up quickly. “And we have duplicates of the pictures right here in our files.” He pulled a thick album from a steel drawer. He thumbed through to the ‘S’ section and opened it in front of Teena and Eddie.

“That’s him!” Eddie said, pointing almost immediately to the picture of the thin-faced man. His name was listed as Harvey Simms. Underneath the photo the man’s job title was typed in a single word—Custodian.

“Now I recognize him,” Mr. Taylor said.

“I’ve seen him working around. A quiet person. The kind you hardly notice.”

164

“That’s the way he would want it to be,” Mr. Ross said.

Teena and Eddie went through the entire book of pictures without recognizing any as the man called Roy Benton. Mr. Ross picked up the telephone and called the Acme Aircraft Company personnel department. He gave Roy Benton's name and the description Eddie and Teena had furnished.

"See if you can get a line on such a person," Mr. Ross instructed over the telephone. "Call me back as soon as you can." He gave the number, and hung up.

"Now, then," Eddie's father picked up the conversation again, "after managing to steal certain blueprints during the week, the men would naturally pick Saturday—their day off—to schedule the pickups by the submarine. We're still assuming, of course, that a submarine actually is being used. It seems the only logical means of getting in and out past our alert Coast Guard. By timing the patrols, they would know when to surface. They would know how long to allow for their divers to row into the bay, get the tube, and return to the sub before the patrol doubled back. It's possible, even, that the submarine carries a small seaplane. After returning to unpatrolled water, they could launch the seaplane to deliver the cylinder to some surface vessel, or possibly to an island or other land base. The submarine itself probably stays around for other pickups."

165

"Those are possibilities," Mr. Ross admitted.

"I mention it," Eddie's father said, "only because, if it's true, the tubes which have been picked up off the sand bar are already delivered. In that case, your blueprints and my radioisotope are no longer secrets. If not, however, both still must be on the submarine. No sub could shuttle back and forth to a foreign shore fast enough to make delivery and get back within a week's time. This is only a guess, but they may lie a few miles offshore during the week as a safety measure and to conserve fuel. They come in and surface just outside the bay each Saturday, under cover of darkness. When they have everything they're after, they'll head home. Since they

166

already have sufficient samples of the isotope, my guess is that they are now after the final blueprints. The small samples of the isotope are now used only as tracers to help locate the submerged cylinders.”

Teena’s father seemed immensely impressed by Mr. Taylor’s reasoning. “It so happens,” he said, “that the blueprints we discovered missing today—added to the others—complete the entire layout of our new secret missile-guidance system. In the hands of an unfriendly nation, there’s no telling to what improper use the guidance system might be put.”

“Then,” Mr. Jamison said, “this must be the end of their assignment—tonight’s delivery of the final blueprints.”

“That’s right,” Eddie’s father said. “That’s how it would appear.”

Mr. Evans came back into the room. “I’ve been arranging a little surprise party,” he said, with a rather tense smile. “I couldn’t help but overhear you, Mr. Taylor, while I was waiting for one of my calls. I think you’ve got that submarine angle pretty well figured out.”

167

“I spent a hitch in the Navy,” Eddie’s father said, smiling. “Operating seagoing vessels—surface or subsurface—falls into a general pattern.”

“True,” Mr. Evans agreed, “and I doubt very much that any submarine refueling tanker would be hanging around even several hundred miles out. Like aircraft traffic, shipping is run pretty well according to schedule. A wandering tanker would simply invite curiosity. But be that as it may, the immediate task is to capture that submarine—if submarine there is. We’re still going on guesses.”

“What do you want us to do?” Mr. Taylor asked.

“It won’t be necessary for any of you to do anything,” the FBI man said. “I’ve lined up all the assistance needed. Everything is set.”

“You’re going to arrest those two men, aren’t you?” Eddie blurted out. “They—they’re traitors!”

168

“They won’t go anyplace,” Mr. Evans assured him. “The important thing right now is that we don’t tip off our plans. Possibly they have various signals worked out with the submarine. Things have to go right on schedule, or we might lose the whole battle. Benton and Simms are small fish and can be landed any time we want. The big thing is the delivery of those blueprints and the isotope. That’s what we’ve got to stop.”

The telephone on Mr. Taylor’s desk rang. “It’s for you, Tom,” he said, handing the instrument to Teena’s father.

“File clerk?” Mr. Ross said, after listening a few seconds. “How about that! Thanks. No, don’t say a word to anyone.” He hung up, and turned to the FBI man. “Well, there’s your Roy Benton. A file clerk. New man. Been at Acme just a little over a month. Can’t figure, though, how he managed to get into the secret blueprint files. They’re kept locked up.”

“Professional spies have ingenious ways of working,” Mr. Evans said. “Anyway, it’s pretty plain now how both the radioisotope and the blueprints happened to disappear. One thing’s equally certain. This is all part of a carefully worked out plan. The job now is to stop that plan—and stop it tonight.”

169

“Oh, I’m frightened,” Teena said. “Spies, and submarines, and—and—”

“Aw, Teena,” Eddie said, “there’s nothing to be afraid of.” Yet he had to clasp his own hands tightly together to keep them

from shaking.

“All right, everybody,” Mr. Evans said, looking at his watch, “within an hour everything will be set up. I’m not free to reveal our plan. However, since you are all involved in this thing, I have no objection to your witnessing the outcome. If an outcome there is. Remember, we’re going primarily on guesses. So, if you want to drive quietly out to the lighthouse, I’ve arranged—”

“Lighthouse!” Eddie exclaimed. “We know Captain Daniels. He’s a good friend of ours.”

“I know,” Mr. Evans said. “I talked to him on the phone. He’s a Coast Guard man, you know. And the Coast Guard is mighty important to tonight’s activity. You might find what goes on out there, and in the bay, extremely interesting to watch.”

170

“Can Teena and I go?” Eddie asked anxiously.

“Of course,” Mr. Evans said. “Without you two, we wouldn’t have a thing to be working on, would we?”

Eddie flushed with pride.

“Of course,” the FBI man went on, “you will have to ask your parents.”

Eddie looked pleadingly at his father. Neither Mr. Taylor nor Mr. Ross voiced any objection.

“All right,” Mr. Evans said, rising, “there’s no time to waste. I’ll see you folks a little later.”

He left the office. The others sat for a moment as though trying to catch their breaths over the rapid developments of

the past hours. Mr. Jamison excused himself to report back to Drake Ridge.

“Tom,” Eddie’s father said finally, “we’d better call our wives and tell them we and the children will be home late.”

171

“Unfinished business,” Teena’s father said thoughtfully.

“That’s right. Unfinished business.”

172



CHAPTER TWELVE

Teena and Eddie, with their fathers, had hamburgers and milk at a roadside stand. As soon as it was dark, they drove toward the lighthouse. They parked the car off the paved four-lane highway which ran several hundred yards back from the rocky point upon which the lighthouse stood. The twisting, twin-rutted road leading to the lighthouse was much more suitable to a jeep than to a modern low-slung car.

They had no more than climbed out of the car, when a uniformed man stepped out of the darkness in front of them. Eddie gasped when he saw the rifle cradled in the stranger's arms, poised ready for instant action.

173

"Halt and identify yourselves!" a voice challenged.

"I'm Steve Taylor from Oceanview College," Eddie's father spoke up quickly. "With me is Mr. Tom Ross. Also our two children."

"All right, sir," the voice said, more pleasantly now. "Been expecting you. Go ahead, sir."

"Wowee," Eddie whispered as they went down the dark road. "I wonder if there are guards all around here."

"Probably," Mr. Ross said. "They certainly set things up fast, didn't they?"

Each time the lighthouse beacon swept around in its circle, it cast a temporary glow upon the road, making walking easy.

When they reached the base of the lighthouse, they noticed several other shadowy forms moving about.

“That you, mates?” Old Captain Daniels stepped out to meet them.

“Hello, Captain Daniels,” Eddie greeted. “It’s us, all right. Our fathers are with us.”

174

Captain Daniels shook hands with the two men. “Quite a party they’re planning out here, isn’t it?” he said, seeming to relish the excitement.

“Apparently,” Eddie’s father said. “Although we don’t know just what they’re planning.”

“You will, you will,” said Captain Daniels. “But right now I’ve got to check my light. Darker’n a ship’s hold with a cargo of tar paper tonight, it is. Won’t be much to see—until things start poppin’. Might be a good idea to sit there in front of my cottage and watch down the coast. Just don’t light any matches, or make undue noise. If you spot any strange lights, things may start happening. I probably won’t see you for a while. Can’t stop the light unless I’m up top.” He turned and started up the spiraling stairs.

“Stop the light?” Eddie said. “I wonder what Cap meant by that?”

“I don’t know,” his father said. “I suppose you can stop those lights from turning in a circle if you want to. Don’t know why anyone would want to, though. After all, the beam is aimed rather high so it can be seen by ships far off the coast.”

175

They waited over an hour. Except for the gentle sound of waves lapping the shore below, and a throbbing Coast Guard plane passing by on its patrol, an eerie silence filled the night. Looking seaward, there was nothing to see but solid

blackness. Three times each minute the beacon from the lighthouse swept a path of white through the sky. Since it was aimed high, the beam didn't touch the water in the bay.

"If anything is happening out there," Eddie said, "how are we going to know about it?"

"I've been wondering that myself," his father replied.

"You'd think they would spot some search-lights along the beach or something," Mr. Ross said.

"They couldn't very well do that, Tom," Mr. Taylor said. "They would risk tipping off the whole trap. That Evans fellow impressed me as knowing what to do. His is a big responsibility, and there certainly wasn't much time to weigh and measure things, but—"

176

"Look!" Teena said suddenly. "Isn't that a light down there?"

Eddie's eyes followed the direction of her outstretched arm.

"It sure is!" he said, dropping his voice to a tense whisper. "It's flashing on and off!"

The light was only a pinpoint in the distance. It flicked off and on in a pattern of dots and dashes which Eddie guessed was some kind of a code. The beam was directed seaward.

"Eddie," Teena said, "isn't that light about where the old fisherman's shack is?"

"I think so," Eddie said, trying to judge in the darkness about how far down the coast the shack was from the lighthouse.

"I figure you're right," a voice spoke behind them. All four turned. They couldn't make out the stranger's face in the darkness, but they could see that he was in Navy uniform. As the lighthouse beacon swung around, Eddie saw lieutenant's

177

bars on his shoulders, and the crossed-anchor insigne of the U.S. Coast Guard on his cap. "Now we'll wait exactly twenty minutes."

"Wait for what?" Eddie asked.

"We're not sure," the officer said. "But in twenty minutes we spring the trap. Might catch some big game, might catch nothing. Please stay right where you are. Keep your voices low. No lights of any kind." He turned and went toward the lighthouse tower.

The Coast Guard officer had just left when Teena grabbed Eddie's arm. "Look!" she whispered, pointing out across the dark bay.

Eddie sucked in his breath as a small light far out on the water flashed three times quickly, then stopped.

"Something's moving into the trap, all right," his father whispered.

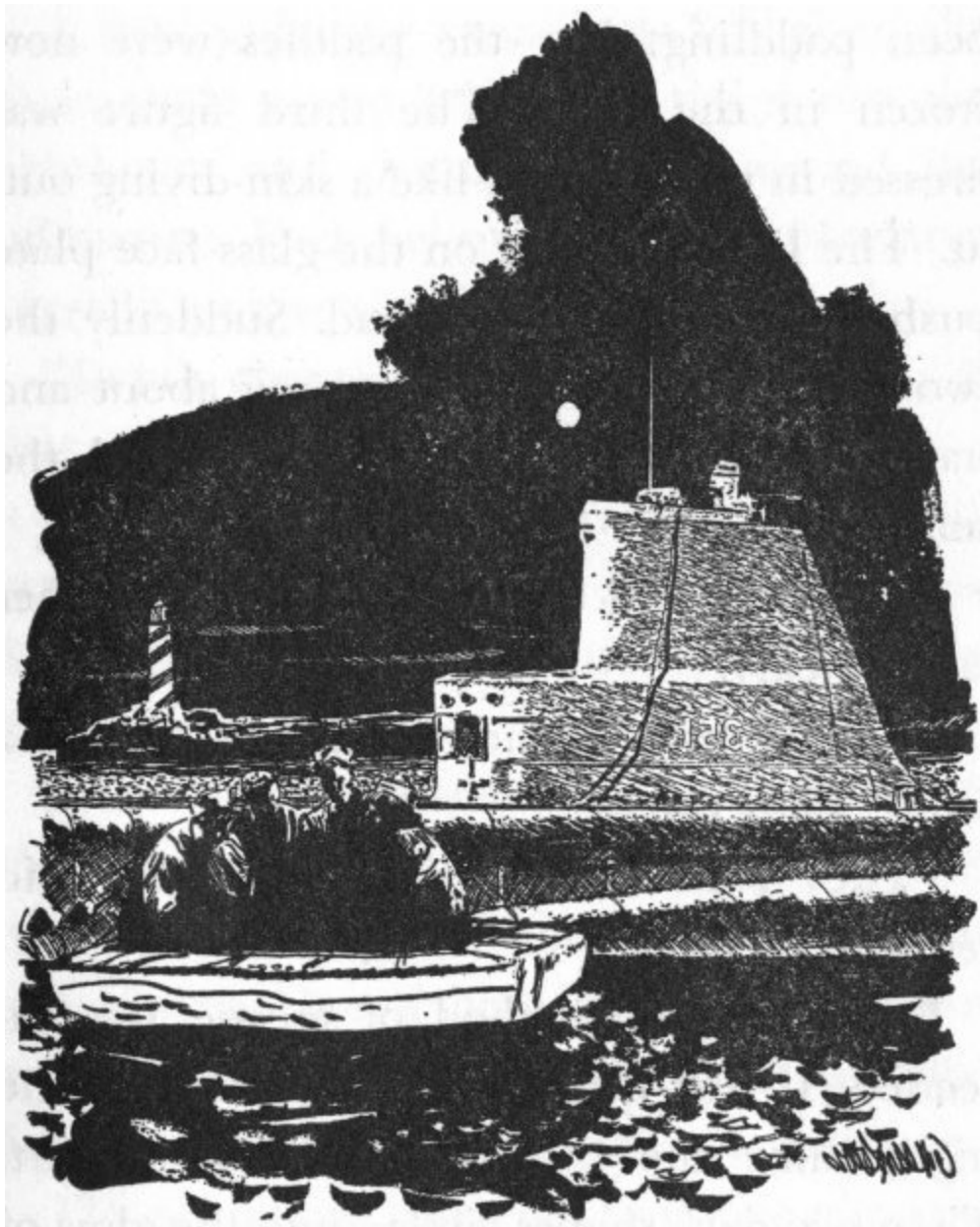
It was unbelievable to Eddie that twenty minutes could be such a long time. No one spoke. Nor were there any other flashes of light to indicate any kind of activity going on. Occasionally, Teena's father consulted the luminous dial of his wrist watch. Eddie wondered if he, too, found that twenty minutes was an awfully long time.

178

Then, as the tension inside of Eddie mounted to the point of bursting, the darkness was shattered by a sudden rush of activity.

It began when the enormous beacon in the lighthouse tower stopped rotating as the beam pointed out across the bay. Then, amazingly, the great finger of light was lowered until it flooded the outer edge of the bay in a brilliant blanket of white.

The sight revealed in the dazzling light caused all four of them to jump to their feet. In the deep water beyond the bay, and approximately half a mile offshore, the deck and superstructure of a submarine stood out plainly on the surface of the calm water. Even at that distance, Eddie could make out the frantic scramble of men pinned in the blinding grip of light.



... he saw the small rubber boat moving in.

180

Then he saw the small rubber boat moving in toward the sand bar of the bay. Three figures were plainly visible in it. Two had been paddling. But the paddles were now frozen in the light. The third figure was dressed in what looked like a skin-diving outfit. The light reflected on the glass face plate pushed up onto his forehead. Suddenly the two men with the paddles swung about and started pulling frantically back toward the submarine.

“That sub will try to dive!” Eddie’s father said quickly.

“But the men in the rubber boat?” Teena said. “They can’t—”

“They’ll be left behind,” Mr. Ross said tensely.

But whatever method of escape was intended, it was quickly blocked. Out of the night came the throbbing roar of aircraft. Then two dark shapes circled into the glow of light from the lighthouse beacon.

“Coast Guard planes!” Mr. Taylor said.

Adding to the brilliance, the Coast Guard aircraft dropped magnesium flares directly over the surfaced submarine, then continued their circling.

181

A new pulsating sound was added to the night scene as two helicopters swept past the lighthouse and slanted directly toward the submarine. Each helicopter carried two large barrellike objects under it.

“Depth charges,” Teena’s father said. “If that submarine tries to dive it’s a goner.”

Apparently the commander of the submarine realized the futility of escape. A white flag caught the light, as someone on

deck began waving it wildly.

More flares blossomed out as the aircraft circled around for the second time. Suddenly two Coast Guard patrol boats nosed into the lighted area. One of them fired a warning shot over the bow of the undersea craft. The white flag began to wave more urgently than ever.

The action had taken less than five minutes. Eddie's mind whirled with excitement. And then, almost as suddenly as it had started, it was over. One of the Coast Guard boats swept into the bay and picked up the men in the rubber raft. The larger boat swerved in and lay alongside the submarine. Eddie could see the crew of the submarine being transferred to the launch. A few remained, while several armed Coast Guardsmen boarded the submarine.

182

Within a few minutes the patrolling aircraft buzzed low over the scene for the last time, then disappeared into the darkness, returning to their base. The helicopters swung back inland. The unused depth charges were still racked securely beneath them.

"Boy, that was some timing," Eddie's father said, as the chop-chop-chop of the helicopters faded into the distance. The submarine had started to move up the coast in the direction of the U.S. Coast Guard depot.

The flares sputtered out, and all was quiet once more on the water. Suddenly the beam from the lighthouse slanted up to its normal position, then began to revolve slowly in its familiar fashion.

"Well, folks," the Coast Guard lieutenant said, coming up behind them, "show's over for tonight."

183

"And a real show it was," Teena's father said admiringly. "That was some display of teamwork."

“All part of our training,” the officer said, obviously pleased the way things had turned out. “We keep the wheels greased. When they have to turn, they turn smoothly.”

“Tonight’s proof positive of that,” Mr. Taylor complimented.

“My instructions,” the young officer said, “are to escort you to the Coast Guard depot. Those taken into custody should be there by the time we arrive. You may have a few questions to ask.”

Teena and Eddie rode with the lieutenant in the gray sedan with “U.S. Coast Guard” printed on its doors. Their fathers followed in Mr. Taylor’s car. Gates opened as they entered the Coast Guard depot a while later. They were escorted into a large briefing room.

Within two hours the investigation was complete enough to draw some firm conclusions. Mr. Evans, of the Federal Bureau of Investigation, and a Captain Foster, of the U.S. Coast Guard, stood before the group. There were many strange faces in the room. Some of the men were in uniform; some were not. Eddie supposed most of them were government or police officials of some kind.

184

Under close guard over to one side of the room were two dozen or more men. They were all strangers. Their uniforms, although of a seaman’s variety, were completely unfamiliar to Eddie.

There were also several men in civilian clothing being held under guard. Among them Eddie saw the two unlucky fishermen he had come to know as Harvey Simms and Roy Benton. They scowled darkly at him. Eddie scowled back.

Mr. Evans seemed to have caught the exchange of glances. “Remember,” he said to Eddie, “I mentioned that we should have no trouble grabbing them whenever we wanted to. Well,

185

we got them. A few others, too. Simms and Benton were sitting outside that old fisherman's shack, still holding the battery lantern they used to signal the submarine."

"Boy, oh, boy!" Eddie exclaimed.

"Now, gentlemen," Mr. Evans went on, turning to the main group, "please regard all that is said here as confidential until it is officially released through the proper channels. If you are wondering why these two young people are sitting in on this hush-hush session, I take great pride and pleasure in informing you that, without their alertness and curiosity over certain suspicious actions, that submarine might now be on its way seaward carrying two secrets very precious to this country's security."

Eddie blushed but felt mighty good. Teena looked at her hands, trying to hide the pleased smile on her lips.

"Mr. Taylor and Mr. Ross," the FBI man said, "this should also please you. We found two more of those sealed metal cylinders inside the submarine."

"Then they hadn't delivered them!" Eddie's father said with obvious relief.

186

"That's right. In fact, the submarine commander has admitted that they have been lying about thirty miles off the coast during the week. Tonight was their third trip to the bay. Incidentally, it was scheduled to be their last. They had plenty of the secret radioisotope, and today's blueprint delivery completed the main set on the new missile-guidance system they also were after. If we hadn't set the trap tonight, we would have been too late—another reason for appreciating the alertness on the parts of your son, Eddie, and Mr. Ross's daughter, Teena.

“Now,” Mr. Evans continued, “we haven’t had time to solve who was behind all of this, or why. We have our ideas, of course, but it’s going to take considerable investigation to draw a full and clear picture. At the moment, I’m not free to reveal to what country that submarine belongs. I did think, though, that you two gentlemen deserved to know that the isotope and the blueprints are safe.”

“It will be a long time, I imagine, before either Mr. Ross or I will hear better news,” Eddie’s father said.

187

“In order not to delay nuclear research, nor to hold up production at Acme Aircraft,” the FBI man said, “we’re sending the tubes with you under armed guard to your laboratory, and you can take over from there.”

“With great pleasure,” Teena’s father said.

“And I believe, gentlemen,” Mr. Evans went on, “that’s probably your main interest at the moment. The rest of it you will doubtless read about in your newspapers within a day or two. I imagine the lights and commotion out around the bay a while ago attracted plenty of attention. Even with the naked eye, it would be simple to identify a submarine lying on the surface. Newsmen are crowding the gates outside right now. They’ll get their story as soon as we’ve filled in a few gaps and get a release from Washington. All I ask is that you do no talking about it until it has been cleared for the press. All right?”

“Of course,” Mr. Taylor said.

“We won’t say a word,” Eddie promised fervently.

188

“No, sir,” Teena backed him up.

“Well, then,” Mr. Ross said, rising, “I guess you won’t need us any more.”

“We’ll call you if there’s anything else,” Mr. Evans said by way of dismissing them. “Before you go, though, I do want to thank you all for your fine co-operation. Particularly you two young people.” He smiled again at Teena and Eddie. “This may sound awfully big—and it is—but you’ve both done a great service for your country. As long as we have alert young Americans like you two, this country’s future is in good hands.”

A burst of applause went up. It seemed a strange place for it, there in the briefing room. Eddie found it almost impossible to conceal the pride that puffed up inside of him. Teena was grinning, too, as they got up to follow their fathers outside.

Both of them took a last look around the room. They saw the group of sullen men in strange uniforms. They saw the tall man and the fat man, whose clumsy efforts at being fishermen had first aroused their suspicions. They saw the pleased looks on the faces of the FBI agent, the Coast Guardsmen, and the others in the “friendly” side of the room. They saw their fathers walking toward the door, carefully carrying the all-important metal tubes.

189

Neither Teena nor Eddie could find anything to say. Then they were outside. The stars blinked overhead. Every few seconds the circling beacon from the distant lighthouse swept its white finger across the sky. The cool breeze from the nearby ocean gave added zest to their high spirits.

“Isn’t it wonderful how everything worked out, Eddie?” Teena said finally. “And to think that we were some help.”

“Yep,” Eddie said. “It all worked out great, didn’t it? Really great.”

Walking proudly, they followed their fathers toward the parking lot.

Transcriber's Notes

- Copyright notice provided as in the original—this e-text is public domain in the country of publication.
- Silently corrected palpable typos; left non-standard spellings and dialect unchanged.
- In the text versions, delimited italics text in underscores (the HTML version reproduces the font form of the printed book.)

*** END OF THE PROJECT GUTENBERG EBOOK ATOM MYSTERY
[YOUNG ATOM DETECTIVE] ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual

work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in

the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work

in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.